HYPERLOOP
GLOBAL

Final Design Review (FDR) Hyperloop Global 2024

POLYLOOP
MONTREAL

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1 General

1.1 About the team

Polyloop is a student organization of 45 people officially founded in August 2020 and associated with Polytechnique Montréal university. Polyloop is the first and only team in Quebec that develops the hyperloop technology. Since the beginning of our operation, we had many great achievements, such as winning the university's best technical society award in 2020. This prize is awarded to the members of Polytechnique Montreal's technical society who have distinguished themselves the most by the excellence of their work and the scope of their achievements. We also participated in the 2022 and 2023 CHC edition, where we had the chance to dismantle our linear induction motor and test our structure on the Waterloo test track. Polyloop also received, at the last two editions of the CHC, the Aeroshells Awards.

Fig. 1: Team picture

Our project for the next Hyperloop competition focuses on several key areas, divided into 7 teams. The external, communication and partnership teams are working on developing Polyloop's business, both in terms of image and relations with the world of industry. As for the mechanical, electrical, physical and civil technical teams, they are working on the capsule as a whole, to finalize the systems presented last year and get everything up and running. For propulsion, we intend to develop a miniature model of a linear induction motor based on the old model presented with new, more realistic objectives. As previously stated, we have undertaken to create a civil team whose projects will be dedicated to designing a test track. This infrastructure will be the focal point of our future developments, enabling Polyloop to imagine, design, build, then test and validate each of its systems individually

and together ahead of the competition.

1.2 Acknowledgements

The physics team would like to express its gratitude towards two individuals who, while not being part of the team, were essential to the progress of the project. Frédéric Sirois is a professor at Polytechnique Montreal who specializes in superconductors and electromagnetism. He is the main reference for motor control techniques and theory. Benjamin Charpentier is a physics master's student at EPFL and a former team member. His experience in simulations is the main pillar of the team's knowledge.

Also, Polyloop would like to extend its gratitude and appreciation to Polytechnique Montreal, Dr. Bruno Blais, Dr. Sami Ammar, Dr. Sébastien Leclaire, Dr. Yousseff Diouane and all the technologists, administrative and support staff of Polytechnique Montreal. Your continued support and guidance is sincerely appreciated.

The admin team would like to sincerely thank all of the team members for their efforts, especially down the final stretch. We were able to accomplish what other schools could not. This is something we should be very proud of, excellent work Polyloop!

Finally, we extend our heartfelt gratitude to all our partners and sponsors whose unwavering support has been instrumental in our journey. Your dedication and commitment have fueled our endeavors and enabled us to reach new heights. Together, we have forged meaningful collaborations and created impactful initiatives that have made a difference in our community and beyond. Thank you for your continued trust, collaboration, and belief in our shared vision. We look forward to achieving even greater milestones together in the future.

1.3 Timeline

In the year 2023-2024, our objectives were centered around the design and construction of a new prototype. To accomplish this, we divided the year into two primary phases: the initial phase focused on the design of the prototype, referred to as the "pod," followed by a phase dedicated to its construction.

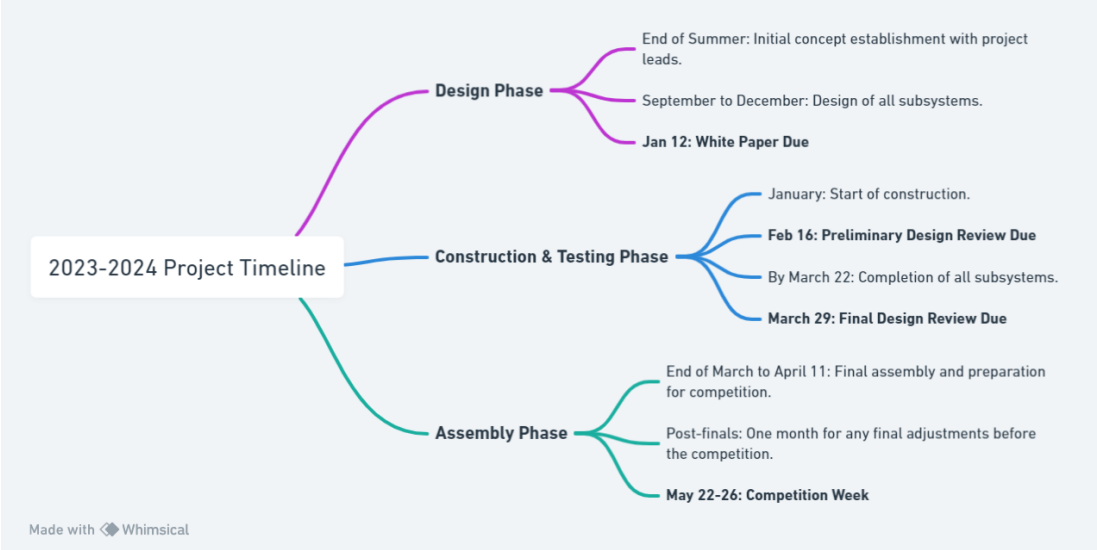


Fig. 2: Polyloop timeline 2023-2024

The design phase launched at the end of the summer, during which the foundational concept or guiding thread for the pod's development was established with the project leads. This stage was crucial for setting a focused direction for the project. From September to December, the project leads and their team, designed all subsystems that would be integrated into the pod. This planning phase was aligned with the competition's guidelines, facilitated by the reception of the rulebook, which allowed for a thorough verification of our design's compliance with competition standards. Any discrepancies identified could be rectified for our initial submission, the white paper. The design deadline was strategically set to coincide with this submission, marking the end of the design phase in early January.

The construction phase was initiated in January, to complete all subsystems by March 22nd. This deadline was intentionally scheduled a week before the submission of the Final Design Package, ensuring sufficient time for subsystem testing. The assembly of these subsystems was planned to span from the end of March until the competition date, with a significant reduction in workload during our finals. This scheduling provided a one-month

window post-finals for final assembly, despite the ambitious nature of integrating all components within such a timeframe. We are confident that with the work made during the winter, this part of the construction phase will be flawless.

Reflecting on the timeline over the year, potential areas for improvement could be identified. An earlier start to the design phase during the summer could have afforded additional time for refining concepts in the autumn. Furthermore, an earlier request for the competition's rulebook from Hyperloop Global would have enhanced our design process, as our strategies heavily relied on the specifications outlined within the rulebook.

1.4 Budget and funding

The financial department is separated in 2 different sections: partnership's team and the treasury.

The partnership's team seek partners to funds Polyloop. Even though Polytechnique university provides the majority of our funding, (about 20 000 $), it is insufficient to complete the prototype. The most significant donation we've receive last fall is a donation of 5 000 $ from Pratt & Whitney. Additionally, we look for suppliers for some expensive commodities that we otherwise cannot afford. This year, we have three suppliers: Golabo who provided us circuit board, Texonic who supplied us with carbon fiber and Oxymax who offered us welding training.

In Polyloop, the treasurer oversees the budget. To dedicate the funds, a technical meeting with the managers of each technical teams to choose which project will be developed for this year. Each team then drafts a budget for their division.

Subsequently, every budget is assembled into a data sheet. Finaly, a function is used to interpolate with our approximate revenue.

1.5 Top-Level design summary

Aiming to compete in the 2024 Hyperloop Global competition, Polyloop was intent on winning as many award as possible. To achieve this, the pod must be lightweight, capable of direction along the track, and capable of accelerating and decelerating in a controlled manner. The pod is a synthesis of six sub-systems: Propulsion, Brakes, Stabilizers, Structural, Power, Embedded and Control Systems.

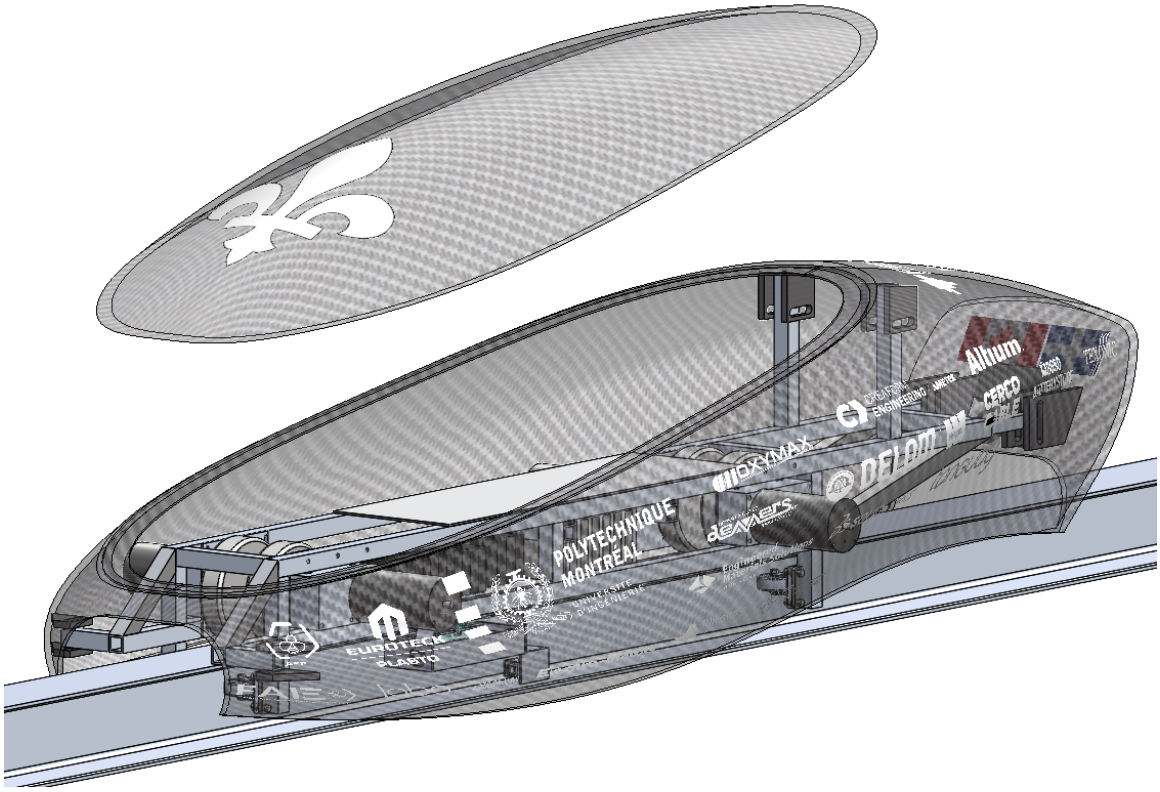


Fig. 3: Overall Pod

2 Subsystems description (design)

2.1 Aeroshell

Overview of the design

After winning the Aeroshells Award two years in a row, we need to maintain our title. We decided to lean toward the original sketch (Fig 4) designed by Polyloop at their founding in 2020. At that time, manufacturing those ideas was too complex, it required composite knowledge.

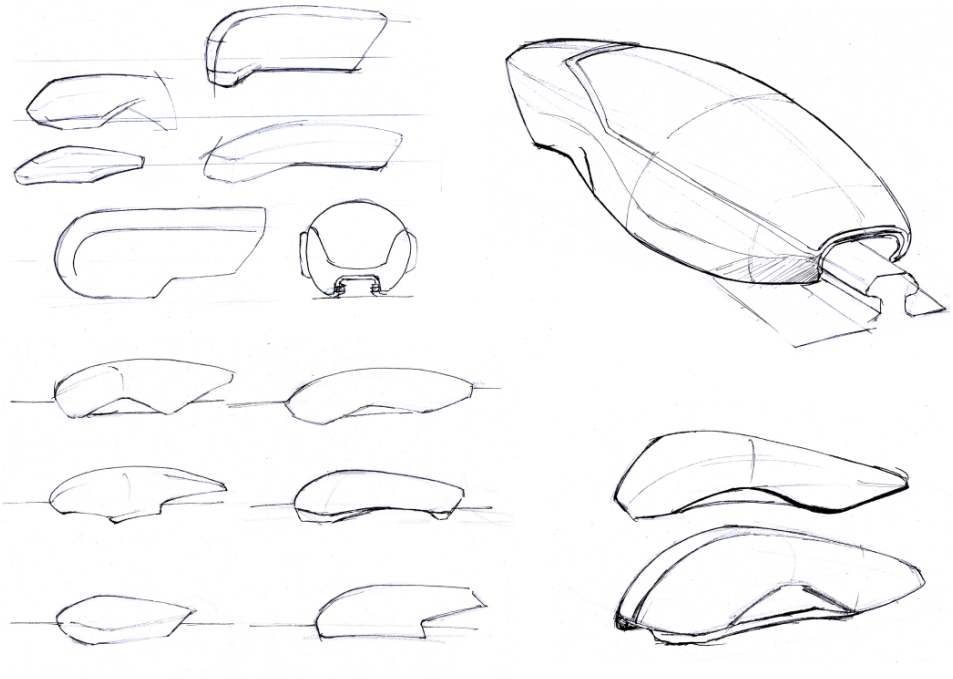


Fig. 4: Exploration drawings

Last year (2023) we launched our first carbon fiber aeroshell. The final product had some issues that will be resolved on this year's pod. First, the last iteration was made out of three parts; two bottom parts and an upper one (Fig 5 and 6). We had problems while trying to fit all those parts together. During the manufacturing process, we did not ensure the fitting between those parts. Both the bottom and the top parts did not have a good finish which made the alignment impossible.

Moreover, it was not practical. Once the upper part is fixed, we could not access the interior of the pod to make some tweaks. Lastly, the final pod rendering was poor quality. The surface epoxy was rough and unacceptable.

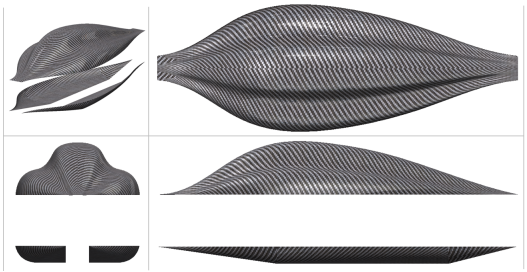


Fig. 5: Last year aeroshell CAD

Fig. 6: Last year aeroshell rendering

Solution

To overcome these problems we decided to design a monocoque aeroshell using the exploration drawings of Fig 4. We will also cut out a hood into this monocoque to ensure the assembly of these two pieces. They will be joined by hood pins to facilitate the manipulation.

Fig. 7: CAD of this year iteration showing
the main body and the hood

It will ensure the pod's protection against wind, dust, and particles. We have also carried out preliminary finite element analysis to conclude that this is not only appealing to the eye but will ensure excellent aerodynamic properties at the maximum operational speed of the pod.

Computational Fluid Dynamics (CFD)

First, we started with modeling the aeroshell in SolidWorks using 3D splines to design the aerodynamic shape (Fig 8). We added other features such as cuts, shell, loft, etc (Fig 7).

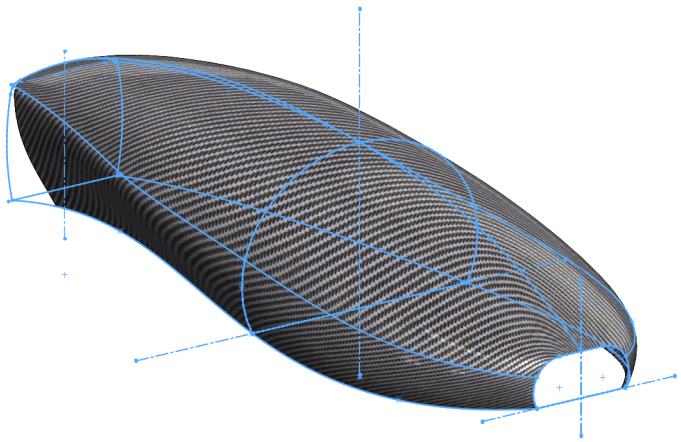


Fig. 8: SolidWorks model resolving the 3D sketch

Then, we used ANSYS Fluent to compute finite element analysis on the aeroshell in order to determine the drag coefficient, the drag force, and the lift and to see the overall behavior of the shell in certain conditions. The air becomes quickly turbulent. Because we have a Reynolds number superior to 10^6 we are in the presence of a turbulent flow. To this conclusion, we use a $k - \epsilon$ turbulence model because it converges more easily and is widely used in industries. It is one of the Reynolds-averaged-Navier–Stokes (RANS) models that is mostly used to simulate mean flow characteristics for turbulent flow conditions. It is a two equations model that gives a general description of turbulence utilizing two transport equations.

Without all the elements inside the pod, we still wanted to represent a realistic situation. We added two rail elements to our aeroshell CFD simulation to see their interaction with the flow. In order to get the desired mesh quality, we had to look at a bunch of parameters. We started by applying a tetrahedrons method to our mesh. Then we tweaked faces and body sizing elements to obtain the right Element quality, Skewness, and Orthogonal Quality (Appendix A.1.1). The mesh with the worst quality is located at each vertical side of the rail. Fortunately, these sections are not studied and their mesh quality doesn't bother the rest. Moreover, the general meshing has a skewness inferior to 0.7.

We also added an Inflation of 16 layers with a growth rate of 20% to well capture the

pod boundary layer (Fig 9 and 10).

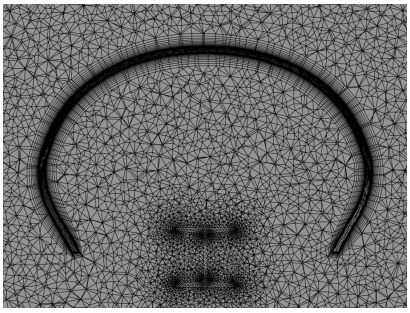


Fig. 9: Cross section of the meshing - distant view

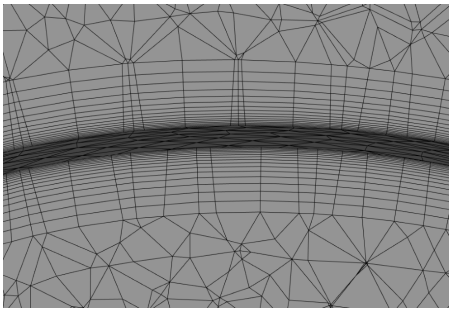


Fig. 10: Cross section of the meshing - close view

After the mesh quality, we needed to analyze the $y+$ dimensionless parameter. On the pod's boudary layer, we obtain $y+$ value between 0.21 and 7.83 (Appendix A.1.2), almost inside the range recommended in the literature ($1 < y+ < 10$). On the other hand, the $y+$ value on the wall with the no-slip condition should be between 30 and 300, and our analysis gives us values $77.54 < y+ < 203.27$ (Appendix A.1.3). This means that the $k-\epsilon$ turbulent model will compute better in the mean flow than in the boundary layer as intended by choosing this model. Although, considering the precision needed, it is enough accurate to consider the results of this analysis. The scaled residuals converged under 0.001 with only 263 iterations using second-order solution methods (Appendix A.1.4). Moreover, the Mach number is relatively low, $M \leq 0.3$. At that number, the Mach number is unimportant compared to the Reynolds number.

For a velocity of $10\,m/s$ we get a drag coefficient $C_d = 0.12904425$(A.1.5), and drag force $F_D = 7.9039601\,N$(A.1.6). Also, the lift force is $F_L = 1.4687748\,N$(A.1.7) given by the lift coefficient of $C_l = 0.023979997$(A.1.8). The gauge pressure is drawn in Fig 11. It reveals that the pressure is pretty constant around the pod. The capsule will ensure good aerodynamic properties during the run while protecting the pod from the environment.

Timeline of the design process

From September to December 2023, we designed and analyzed the aeroshell shown in figure 7. We started machining small prototypes in foam to eventually machine a mold to apply carbon fiber in 2024. Meanwhile, in January 2024 we plan to test vacuum bagging carbon fiber on smaller prototypes. Moreover, we are in discussion with technicians to test one of those prototypes in our school wind tunnel. Then we will build the final full-scale prototype.

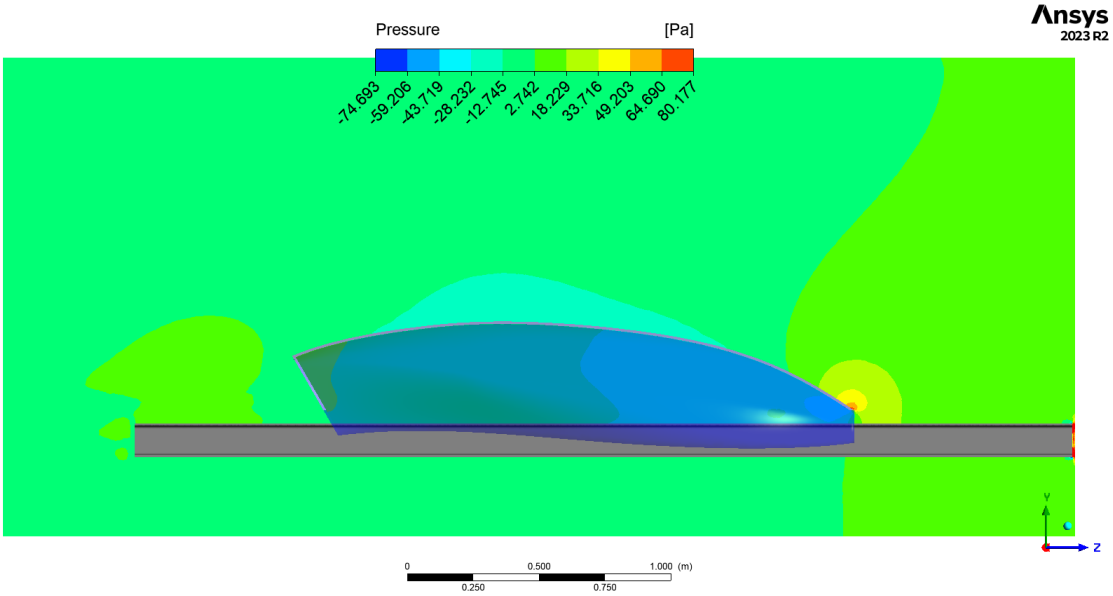


Fig. 11: Gauge pressure around the pod

Challenges met

We faced a wall when we had to find a way to fix the aeroshell to the frame. We successfully found a clever way to use a swivel ball joint with a flat base mounted inside the pod. Another challenge concerns the calculation time of CFD analysis. To have precise results, you need an excellent mesh, but in exchange, it requires more calculations. After a simulation, we changed our design and had to recalculate the analysis again and again. This could take up to 24 hours.

2.2 Structures/Chassis

Overview of the design

This year again, our team will design its pod's structure upon structural design concepts. It will carry previous hyperloop structure features such as a frame made out of 1-inch square aluminum tube.

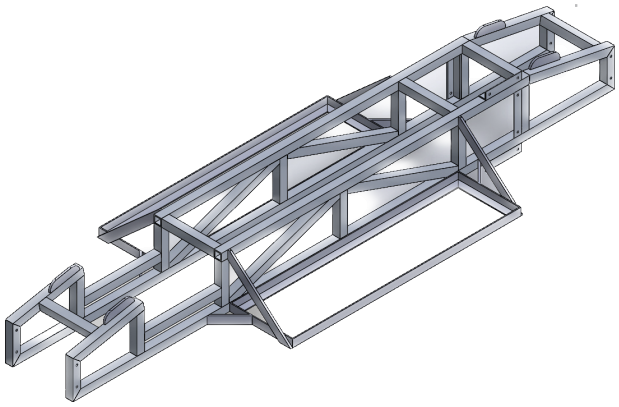


Fig. 12: Typical aluminium structure

Structural design is a methodical investigation to get the economical specification of a structure or a structural element to carry the predicted load safely. With the application of structural design, we can obtain the required size, grade, reinforcement etc. of structural members to withstand the internal forces calculated from the structural analysis.

Modelling

That being said, we start by modeling the desired structure's skeleton, such as its foundations, frames, and other elements. It's essential that it can anticipate loads since it would be the basis for the whole process. We need to consider the max mass of 236 kg and the max length of 3 meters for the main pod. It will be designed symmetrically over the rail to have a center of mass that lies on the track. The second pod will be much smaller and lighter, so structural analysis will not be necessary. However, the main features of the main pod will help design the other one.

Load Analysis

Next, one must identify all of the possible loads that the structure may experience during its lifetime. Moreover, these loads are actions that threaten the structure to serve its

purpose. Usually, we refer to forces, torques, pressure, etc. In real life, different loads act simultaneously with the structure; hence, to accommodate this instance, load analysis also involves creating load combinations depending on the standards used. For instance, a simultaneous action between dead load and live load can be a load combination.

Our structure is first loaded under its own weight. Then, the acceleration applies another set of forces. There are also friction and drag forces opposing the movement of the pod. Finally, the vibrations along the ride could bring the structure to resonance. The forces acting upon the pod are referred to in the table 1. Also, we need to consider critical situations such as sudden stops from top speed.

Tab. 1: Forces acting on the main pod

Force	Details	Value
Pod Weight	$100\,kg$ (All component included)	$981\,N$
Pod Acceleration	$0.93\,m\,s^{-1}$ (Assuming constant acceleration)	$93\,N$
Gearbox Friction	Force generated	$39.763\,N$
Pod Drag Force	$10\,m/s$ (Compute from ANSYS fluent)	$13.14\,N$
Sudden Stop	Conservation of Momentum	$9 \times 10^5\,N$

Structural Analysis

After identifying such loads comes structural analysis. The goal here is to investigate how our model behaves with the different load combinations made. There are different methods to analyze a structure and would depend on a lot of factors. A full analysis of a structure usually entails identifying the internal effects of different load combinations (normal, shear, moment, and torsion stresses), and the deformation/deflection of the structure. As a result, the observed effects would be the basis for designing the individual elements of the structure. We then use ANSYS to calculate the various load effects to assure a minimal safety factor of 2.

Design

Structural design is, perhaps, the most important step in the process. It involves dimensioning the different parts of the structure based on the results of the analysis. Also, it is concerned with choosing the right orientation, section, and other details that are relevant to the structure. For proper design, it would depend on the philosophy used (strength-based or performance-based).

We will use 1-inch square aluminum tube 6061 T6 in this frame design because it is much lighter than other metals. It has very high mechanical strength levels and that is why it is so broadly used in aeronautics, a very demanding field. Also, it is very resistant to many forms of corrosion and has almost zero maintenance cost. It also has the property of not being ferromagnetic, an important feature considering the use of magnetic fields in our overall design. The whole is also welded with 4043 aluminum wire .035" in diameter to solidify the assembly. Welds are stronger than nuts and bolts or rivets and it also doesn't loosen with vibration. The tube tichness is set at 1/8 inch considering our welding equipment and our welding experience. Welding thinner than 1/8 inch becomes harder with our equipment. However, other components added to the chassis will be firmly fixed to the aluminum chassis.

An initial modal analysis revealed flaws in our design. Both ends of the chassis were loose and fully distorted at different vibrating frequencies. The eighth mode is shown in Figure 13 at $48.209\,Hz$. This design is thus non-viable.

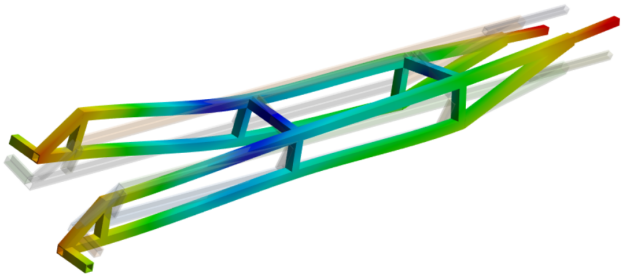


Fig. 13: Not to scale chassis deformation representation at $48.209\,Hz$

We have modified this design several times to achieve a frame that will not distort as much as an equivalent frequency. We added a rod on both ends to counteract the displacements observed in the figure 13. At a frequency of $43.835\,Hz$ we observe a max displacement of $16\,mm$ at the red spot in figure 14 compared to $837.45\,mm$ in figure 13. Thereby, the structure will resist the vibration and the different mode encountered. The main results of the modal analysis are presented in appendix A.2.1.

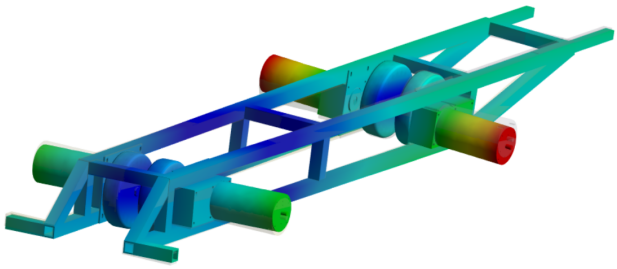


Fig. 14: Not to scale chassis deformation representation at $43.835\,Hz$

Following the dynamic analysis, we apply static load to conclude on the rigidity. We first test the situation where two people lift the pod at both ends. The gravity exerts a certain force. We also add another $30\,kg$ of material on top (general and over estimated). The maximal deformation is about $0.38137\,mm$ and shown in figure 15 using a 27 scale factor. The equivalent stress at this spot is $9.3981\,MPa$. The safety factor is well over two in that case.

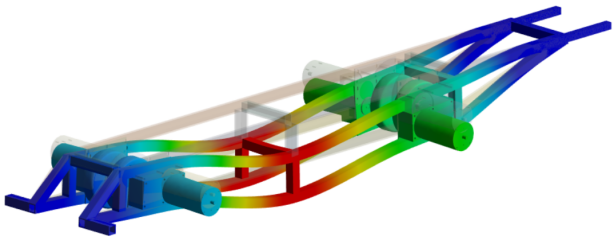


Fig. 15: Deformation under gravity (27 scale factor)

The last case scenario refers to a sudden stop from max speed. Pretend the pod hit an obstacle sticking out of the track causing a sudden stop. The front of the pod will feel a force equivalent to $914\,000\,N$. The max strain feel by the pod will be around $7\,742.9\,MPa$ and the maximum deformation is $100.35\,mm$ (Fig 16). We conclude that the structure won't survive such an impact. At least there would be permanent damage.

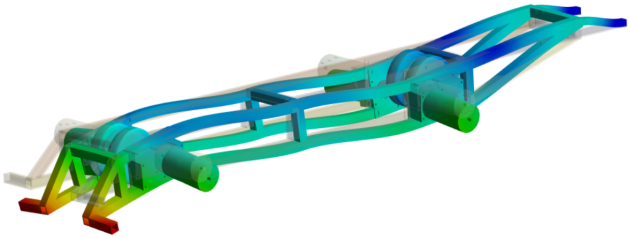


Fig. 16: Deformation under sudden stop ($914\,000\,N$ of force - true scale)

To conclude, the structure resists easily under its own weight and more. It will certainly resist the pod's acceleration ($0.93\,m/s^2$). Moreover, the added rod helps counteract the vibration. We are not designing a tank we don't want to resist to exponential forces, it will most certainly not resist a full-speed impact.

Timeline of the design process

We are currently finalizing the design phase. We have spent the last months working mainly on other subsystems. In the past few weeks, we did the modeling and the load analysis. We did a first SolidWorks model and performed a structural analysis using Ansys identifying

the internal effects of different load combinations. We are currently working on various configurations to explore the best possible outcome. From there we'll do welding tests and then build the final product.

Faced challenges

The first challenge we faced was a timeline issue. We did not have time to carry out a complete analysis of other frame material possibilities. We used the aluminum tubing concept developed for the last two pod iterations.

To go lighter using the same material, we would need to change the fixing method. Our welding equipment doesn't allow us to weld thinner than 1/8". We should also limit the free space in the pod. We designed it to be able to gradually include all subsystems separately. Indeed, each subsystem has more space than required to ensure it fits.

For the next iteration, we should redesign entirely the structure. Investigate and perhaps choose another material, one that fits all our requirements, and plan more precisely where all the components will be.

2.3 Guidance

Lateral guidance

The lateral guidance and suspension system (Fig. 17) consists of four fixed wheels positioned on both ends of the frame and on either side of the track. The 50 mm wide tires are made of high-quality solid rubber (hardness 80 shore A) which will ensure both surface preservation of the track and smooth rolling. The assembly also includes Belleville washers and rubber washers to absorb vibrations during operation.

By trial and error, we found the ideal number of rubber washers to put on each bolt to obtain a good mechanical fit with the track.

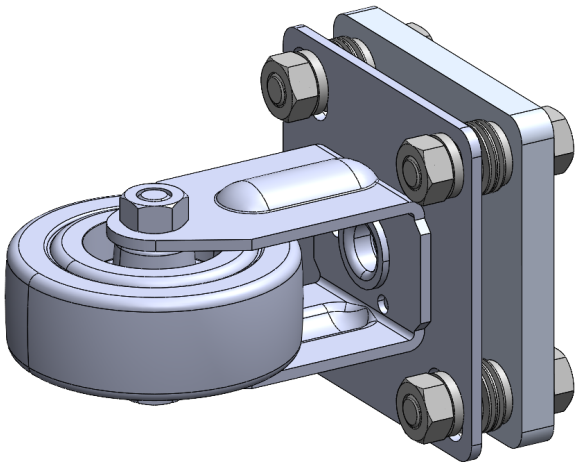


Fig. 17: Lateral guidance and suspension

The lateral suspension system uses a rubber washer to absorb the vibrations of the pod. They are only useful in compression; they don't have any influence on the pod when they are in traction. Thus, with CES Edupack, the compression properties of the material that constitutes the washer have been evaluated.

Stiffness of the spring

In the next few formulas, a unit factor will be used between the movement of the wheel and the spring. To respond to this uncertainty, a security factor of three will be used. To find the necessary spring stiffness, two factors are important: the energy needed for a smooth movement of the capsule and the force needed to realign the pod. For the first factor, we suppose a conservation of the angular moment:

$$I\omega_i + mv_i(r-h) = I\omega_f + mv_f(r) \tag{ii}$$

Where v_i and ω_i are the initial linear and angular speed of the wheel, v_f and ω_f are the final linear and angular speed of the wheel, m is the mass of the main pod, r is the radius of the wheel, h is the max misalignment possible ($10mm$) and I is the inertial moment of the wheel ($I = \frac{1}{2}mr^2$ for a disk)

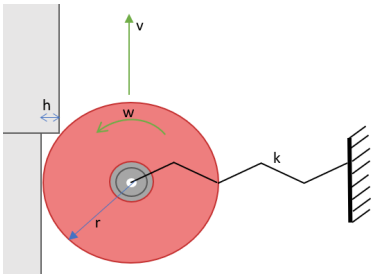


Fig. 18: schematic

With (ii) and assuming an initial speed of $30\,km/h$, a radius of $25\,mm$ and a mass of $100\,kg$, it is intuitive to find the final velocity of the wheel:

$$v_f = 25\,km/h$$

With this new value, with the energy conservation law, it is possible to find the minimum stiffness k needed to move up the wheel without the wheel jumping out of the track with a security factor of 3, the stiffness needed for such a speed differential is $k = 127\,000\,N/m$.

$$\frac{1}{2}mv_i^2 + \frac{1}{2}k\Delta x^2 = \frac{1}{2}mv_f^2$$

For the second factor, Since the spring need to be at half its maximal deformation when it is stable and we don't want it to be completely compressed the force of the spring at $\frac{3}{4}$ of its travel should be stronger than half the friction force plus the force of the spring on the other side of the track which should be at 1/4 of its travel.

$$\begin{aligned}
\frac{3}{4}k\Delta x &= \frac{1}{4}k\Delta x + \frac{1}{2}(mg\mu) \\
\frac{1}{2}k\Delta x &= \frac{1}{2}(mg\mu) \\
k\Delta x &= mg\mu
\end{aligned}$$

Since we need to know the maximal travel of the spring, we can't have the value of the constant needed to realign the pod.

The washers are made of polyester fiber, most specifically of SBR unreinforced. This material as the property of a compression set of 5% to 10% at $23°C$ and 10% to 20% at $70°C$. This means that for lateral suspension, at $23°C$ the maximum compression would be 1.27 mm and 2.54 mm at $70°C$. Knowing that the temperature in the pod would be between $23°C$ and $70°C$, the rubber will be perfectly able to resist the vibrations. We then stack washer to obtain our desired stiffness.

Vertical guidance

The set of 4 propulsion wheels discussed in section 2.4 is used as the vertical guidance. Indeed, the pod will not levitate, thus there is only a down force to counter due to gravity. The set of wheelmotor are equally spaced in pairs (Fig 19), thus sharing the load. This configuration assures stability.

2.4 Propulsion / Levitation

The pod will be powered by DC electric motors mounted on 6 inches wheels proportioned by a speed reducer to improve the torque, thus the acceleration (Fig 20). We thought about doing an hybrid propulsion using both friction wheel and contactless propulsion given by a Linear Induction Motor LIM. But the LIM ratio weight-thrust is too high to be added onto the main capsule. It won't help the propulsion, in fact, it might be a liability. That is why we are building a pod with DC motor and also working in parallel on a custom Linear Induction Motor.

Propulsion of the pod

The electric propulsion system works by friction. In fact, we opted for the use of wheels in contact with the rail propelled by DC brushed motors. Compared to a previous iteration, which had only one wheel powered by two DC motors, this propulsion is achieved by the arrangement of two pairs assembled in two overall. The choice of this configuration is based on the principle of stability, which from a practical standpoint refers to pairs. The mass of the capsule is thus distributed across the four motors, reducing the effort provided per motor.

Secondly, stability is achieved through the maximum spacing between two aligned wheels. By increasing the distance between two support points, stability is increased. The motors of the same pair are not connected axially, allowing each to react to track imperfections and vibrations independently to the other.

The choice of motors for this iteration is based on the energy calculations. The required power and torque were calculated for a two, four, six, and even eight motors configuration. According to the specifications of the different engines, our choice stops on the E30-400-48-G4. According to the manufacturer's performance tables (appendix A.3.1), the motor chosen has a torque of $4.77\,N \cdot m$ for $16\,amps$, which allows us to calculate the acceleration of the capsule by the sum of the moments at the point C (appendix A.3.2).

In an effort to minimize mass and go as fast as possible within the confines of the track, four DC brushed motors with a ¼ speed gearbox were chosen (Fig 19). The choice of this gearbox was based on an analysis of the distance traveled before reaching the motors' maximum speed. Increasing the reduction ratio certainly decreases the outgoing speed, but it increases the torque and therefore the acceleration. The current configuration of four DC brushed motors with ¼ gearbox as well as a six inch wheels, would allow one to go at a theoretical speed of $9.14\,m/s$ within the limits of the track.

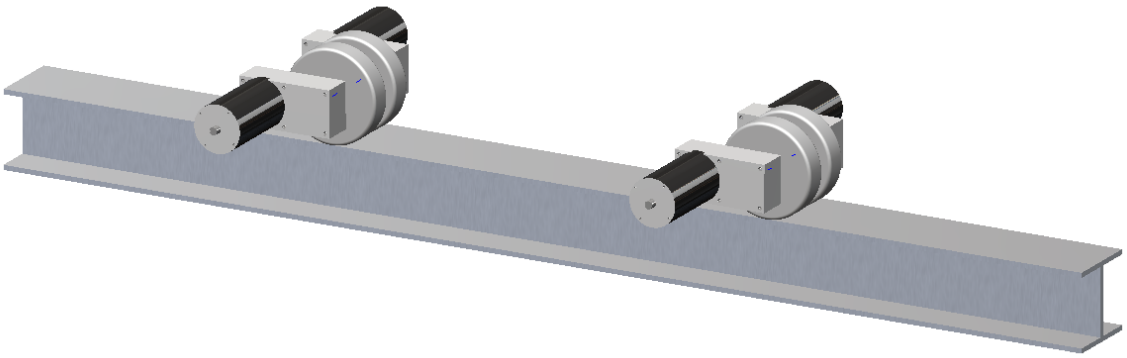


Fig. 19: Assembly of four gear-motor-wheel set in two pairs symmetrical onto the rail

Following the advice of different teachers, we bought an already existing set of motors, gearbox, and wheel. The reasons given by these professors included the complexity of achieving precision in constructing such a system.. We, therefore, purchased this assembly from *AmpFlow* (Fig 20), a manufacturer that offers quality machined products. An aluminum, machined gear box enclosure will certainly prevent part from dislocating in the event of failure.

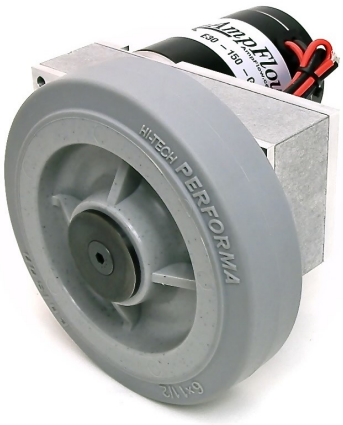


Fig. 20: 6 inches wheel on GearMotor

The reducer will be fixed to the chassis through a 1/8" aluminum plate. There will be a layer of lubricant between the aluminum plate and the square aluminum tube. The screws will go through a straight slot which will allow slight up-and-down movement. The displacement is absorbed by the rubber bumper pad on both sides of the reducer laying on the aluminum structure. It will ensure stability in the face of track irregularities. All the

fasteners will be in good condition and properly installed.

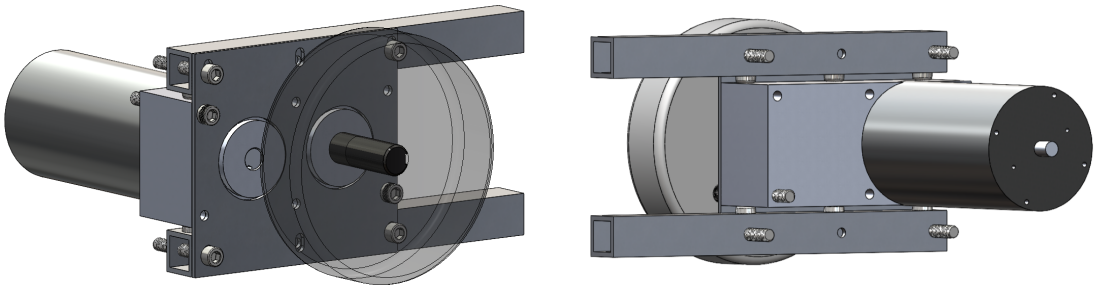


Fig. 21: Representation of the fixing methode of the speed reducer

Analysis

Also, the team did a measuring test to evaluate the impedance of each motor and see if the error was under 10%. Otherwise, the team must integrate a voltage synchronizer. The results came back positive with an oscillation from $190 - 200\mu H$. To continue, we verified if the vibrations coming from the nominal power were being tolerated by the motors. Fortunately, since the reducers are machined and the wheels are molded, their vibrations are minimized and their performance is maximized. Therefore, these tests were concluded.

Moreover, those speed reducers will be used as brakes as described in section 2.6.

Levitation

Unfortunately, this year we still propel the pod using friction, thus no levitation. However, our physics team is working on a new LIM design. They we'll be able to test it and perform demonstration for proof of concept.

2.5 Cooling Systems

Context Setting

The recent competitions revealed an issue with the circulation of heat flow of the pod. Indeed, exposure to direct sunlight at its zenith in temperatures exceeding 30 degrees Celsius weakened the external structure. Our structure is made of carbon fiber composite solidified by epoxy. Some areas on the structure had softened due to the degradation of the optimal properties of the carbon fiber and epoxy mixture. Additionally, the heat absorbed by the shell was such that handling it could cause mild burns to the fingers. Also, since our technical team does not yet possess mastery of a Linear Induction Motor (LIM) requiring 25,000 W of electrical power, the team will introduce four direct current (DC) rotary motors on wheels to substitute for the main propulsion system until the LIM technology is mastered.

These DC motors are connected to speed reducers to increase torque. Recently, a short no-load test on this arrangement revealed an operating temperature of over 50 Celsius degrees for the reducers. This thermal gain over a short period raises questions about the need of an internal cooling system. Therefore, a reassessment of the current material and/or coating of the external structure, as well as the implementation of a cooling system, is to be expected. Cooling systems of the pod will be separated into two big sections which are: Cooling of the speed reducers and Cooling of the aeroshell.

Cooling of speed reducers

Proposed Methodology :
1. The team will assess the current issue:
a) Evaluate the heat-related problems on various components of the capsule.
b)Examine the impact of heat on the aluminum of the speed reducer.
c)Determine material and environmental characteristics.

2. The team will identify the characteristics of current materials and the environment (conductivity, absorptivity, conduction coefficient, convection coefficient, etc.). The key will be to formulate appropriate assumptions regarding initial temperatures.

3. The team will then conduct Heat Transfer Analysis using EES (Engineering Equation Solver). The heat transfer will be evaluated on the speed reducers to determine heat transfer characteristics.

4. From the results of the previous step, the team will propose cooling solutions based on

preliminary analysis. An example of a solution could be as simple as adding a fin system that uses incoming airflow to cool down the pod.

5. When the solution is chosen, the team will validate numerically with EES that the chosen solution effectively regulates the system temperature.We will also extend the analysis to a 3D Model so that we can finally use Ansys Workbench and Steady-State Thermal Function.

Cooling of aeroshell

Proposed Methodology
1. The team will assess the current issue :
a)Evaluate heat-related issues on various components of the capsule.
b)Determine the glass transition temperature (Tg) of epoxy.
c)Determine material and environmental characteristics.

2. The team will identify the characteristics of current materials and the environment (conductivity, absorptivity, conduction coefficient, convection coefficient, etc.). Again, the key will be to formulate appropriate assumptions regarding initial temperatures.

3. Then, the team will conduct Heat Transfer Analysis using the EES (Engineering Equation Solver) of the shell considering radiation and convection. From the heat transfer results, the team will propose solutions based on preliminary analysis such as introducing a new material with lower absorption and higher refractoriness.

4. The team will reanalyze the chosen solution with EES to validate.

5. Team will validate numerically with EES that the chosen solution effectively regulates the system temperature. We will also extend the analysis to a 3D Model so that we can finally use Ansys Workbench and Steady-State Thermal Function.

Conclusion

It should be noted that the solutions chosen for these two cooling systems will be studied and presented during the current semester as part of the capstone project of two members of the mechanical team. The chosen solution could be integrated into the final module depending on the progress of these two synthesis projects.

2.6 Braking

Following friction brakes on last year pod, our team want to improve its braking system. Having the frictionless hyperloop concept in mind, we lean toward a magnetic brake system. This option ensures the safety of the event while not damaging the track. Those brakes are not in contact with the rails, thus no scratch or damage could be induced. In fact, no instantaneous pressure of more than 34 MPa at any point. Moreover, because it is distant to the beam web, it will be tolerant of all potential variations and defects in the track.

Many possibilities are available. For starters, we need to choose between permanent magnets and electromagnets. Then we need to analyze different acceptable braking configurations and finally demonstrate that they will not overheat the track past 70 °C.

Permanent magnets or electromagnets

One of the many challenges of designing a magnetic braking system is to make sure the magnetic field of the magnet is close enough to the conducting material. This ensures the induced current is properly generated and the braking force generated is enough to slow down the pod. Each proposition comes with their advantages and disadvantages.

Electromagnets First and foremost, an electromagnet has an electromagnetic field that depends on the current flowing inside it, which means it can be adjusted to our needs. It is also less strict with the distance required between it and the rail due to the variable magnetic field. However, it also comes with potentially grand needs for power. The power could either be shared with the batteries of other subsystems or it could have its own set of batteries. Sharing power with another subsystem could inhibit that subsystem, making it inefficient. Having its own battery set will add some mass to the pod and could make it harder to move.

Permanent magnets Furthermore, permanent magnets have, unlike an electromagnet, a fixed magnetic field, meaning the distance between it and the rail could be more restrictive and it could still affect the trajectory of the pod even if it's some distance from it. The challenge here is to move the magnet close enough to the rail to induce enough current and keep it from affecting the pod by moving it further from the rail or completely block the magnetic field from reaching the rail. There is a way to increase the intensity and the reaching distance of the magnetic field of permanent magnets using a Halbach array. A Halbach array is composed of five magnets in a specific configuration as shown in Fig. 22. Their magnetic field is all aligned in a specific way. This configuration makes the magnetic

field more intense on one side and almost nonexistent on the other. It allows a greater gap between the array of magnets and the rail without losing any braking power. Still, this solution doesn't solve everything because we still have to move the magnets even further than before or block the magnetic field.

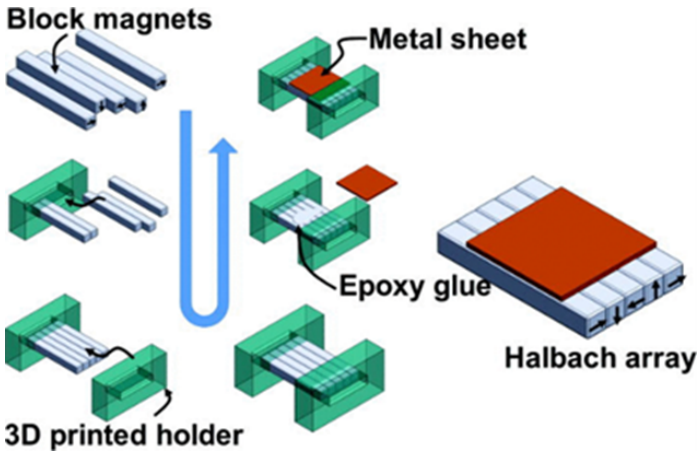


Fig. 22: Halbach array permanent magnet configuration

Analysis of Halbach array

To reinforce our understanding of magnetic brakes, we ran simulations using COMSOL. Our goal was to get a better comprehension of a Halbach array and have some data to support our final decision. We started by trying to mimic the exact set up we would have when it will be build, shown in appendix A.4.1 in Fig. 78. The configuration shown is optimal, it is the one that puts the brakes to what we can theoretically do with our current means, about $5\,mm$ from the rail. There is only one part of the brake because we wanted to get a better understanding first, rather than have even more components and no understanding at all. A more realistic approach is scheduled. The simulation results are dependent on the velocity which has a range of $0\,m/s$ to $30\,m/s$. Fig. 79 in appendix A.4.1 shows a peak around $6\,m/s$, and then the braking force suddenly lowers as expected. It is optimal for our pod as the highest braking forces are in our range of speed. One thing to note is the values of the forces in this graph are in N/m. We simply need to multiply each value by the width of the Halbach array to get the actual braking force. For instance, the highest braking value obtained is $26\,777\,N/m$. Multiplying by $0.03\,m$, the width of the Halbach array, gives a braking force of $803\,N$.

Finally, our team decided to choose the Halbach array with permanent magnets. This solution requires much less power from batteries, which was the main issue of the electromagnet. We also have data from simulations to backup our decision.

Braking configuration

The Halbach array requires a total of five magnets as mentioned earlier. As shown in the Rulebook, the arrangement of the braking system must be symmetrical, meaning we must use a minimum of two Halbach arrays, so a total of 10 permanent magnets. The ones used in our system are neodymium N52 magnets. Those can be quite expensive so we can't use as many as we want. This limits us in the configuration but it isn't an issue.

The five magnets are touching each other, shown in Fig. 23, to maximize the efficiency of the Halbach array. Usually, a Halbach array is made of squared magnets. However, in our case, the ones that we have are rectangular. This changes the resulting braking force. It is an insignificant difference in our case. They are fixated on an aluminum plate with aluminum bolts which makes the handling easier as aluminum is non-magnetic. The complete assembly is shown in Fig. 24. The system has a lock to maintain it downward while we don't need to brake. We can see it in Fig. 25. It retracts to let the array go up to brake. The system is failsafe as the lock will retract automatically when there is no more current in it.

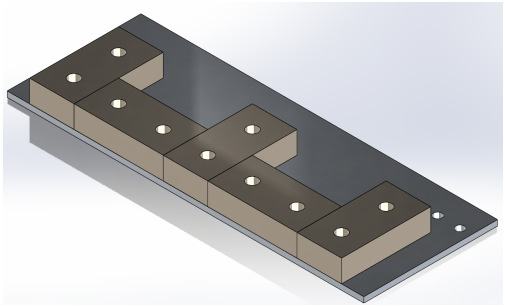


Fig. 23: Design for the Halbach array arrangement with completely visible magnets

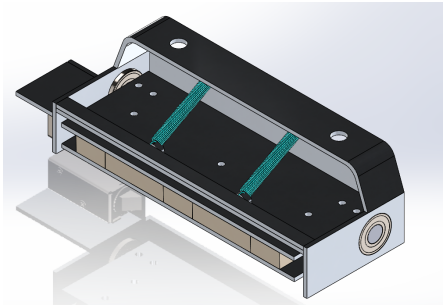


Fig. 24: Complete design for the braking system

We decided to put one Halbach array on each sides of the track shown. Fig. 25 shows one half of the placement.

Challenges met

During the conception of the magnetic brake system, the team dedicated to the magnetic brakes had very little knowledge on the matter. Also, it is very uncommon for a magnetic brake system to take the form shown in Fig. 25. Usually, there is a fixed rotating disc that is used as the conducting element in the magnetic brake system, usually in heavy vehicles. Having close to nothing to start with was proven quite challenging, but as seen earlier, we've found a functioning design. Also, levitating land vehicles aren't common, which makes their braking technology quite inaccessible, making it harder to design our brakes. Another

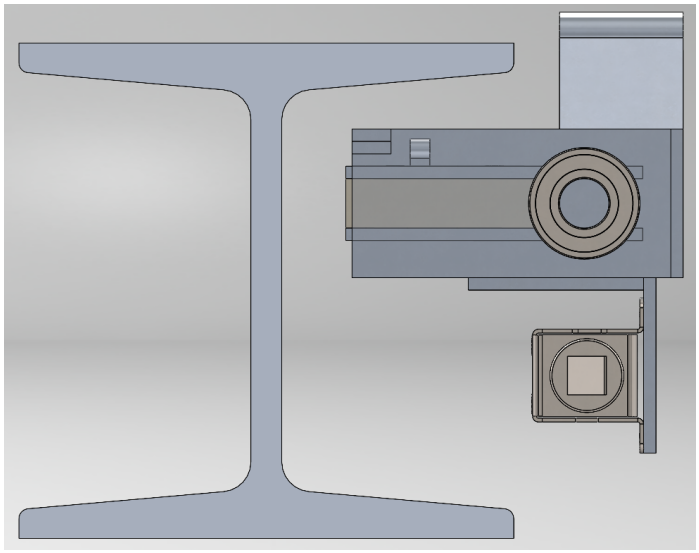


Fig. 25: Placement of one half of the braking system

challenge was to choose between the electromagnet and the permanent magnet, as they both have interesting characteristics. It took a couple of months of reading and debating to choose the permanent magnets.

Timeline of the design process

Back in September, the different mandates were assigned to the different members of the mechanic team and one of the was the conception of magnetic brakes. Although some ideas and prototypes were there before, making it a reality and implementing it in the pod was something never seen before in Polyloop. It was only in late December that we made the decision to stick with permanent magnets due to the results of the simulations and the knowledge we got from them. The design shown in Fig. 25 was produced some time afterwards. We still have to find a way to nullify the effect of the magnets. Then, there will be a testing phase that is to come after building the brake. Afterward, they should be operational.

Emergency brake

Although the magnetic brake should act as a fail safe system, our speed reducer can also bring the pod to stop. Our motors are connected to Standard Speed Reducers shown in figure 26. A single stage of roller chain reduction is generally considered to be about 98% efficient. Add in some friction from the five bearings, and we are at about 90% efficiency.

Fig. 26: Opened Ampflow standard gear speed reducer

The friction in the speed reducer is about $40.99\,N$, thus, the total friction force is $F_f = 163.96\,N$. In case of total power failure, it will be capable of stopping the pod. Indeed, the pod wheels are always in contact with the rails and always apply the same friction force. It also brings the pod to a complete stop after the magnetic brake effect. Those magnetic brake has a braking force proportional to the relative speed, thus it can't theoretically stop the pod.

We obtained the friction force by testing the motor. We rolled a string around the wheel to which we hung a mass. Then, with an high-speed camera, we determined the time the mass traveled a certain distance. We then summed the force to obtain the friction force for a single-speed reducer. From the Appendix A.4.2, we calculate the deceleration and the time to immobilization. From there we can calculate the approximative distance traveled by the pod in case of total failure. Indeed, by using simple kinematic equations and hypothesis, we can get the distance traveled with quantities associated with the motion of objects featured in Tab 2. With the calculation in appendix A.4.3, the pod will travel $26.27\,m$ before complete immobilization in total power failure.

Tab. 2: Quantities associated with the motion of the pod

Variable	Value
Deceleration Time	$\Delta t = 5.7464\,s$
Nominal Speed	$v_0 = 9.14\,m/s$
Final Speed	$v_1 = 0\,m/s$
Deceleration	$a = -0.409\,m/s^2$

2.7 Power Systems

For the entire power supply system, we will use lithium-ion cells; Samsung INR18650-35E for the high-voltage battery and Samsung INR21700-50E for the low-voltage battery. The power supply system is divided into two different systems: the high-voltage system, which consists of 4 batteries (13s2p) of 48V and 16 A continuous and 26 A peak each, and the low-voltage system, which consists of a battery (7s5p) of 24V and 19.6A continuous and 29.4A peak. Thus, we have complete independence between each block of the power supply system. Each high-voltage battery block has a theoretical usage autonomy of about 25 minutes, while the low-voltage battery has a theoretical usage autonomy of about 42 minutes. For the bus bars, we have chosen a nickel, steel and copper alloy from our partner SIGMACLAD. The advantage of such technology lies in the ability to have bus bars with excellent weldability, strength, and high conductivity.

Item	Specification
3.1 Standard discharge Capacity	Min 3,350mAh - Charge : 0.5C(1,700mA), 4.2V, 0.02C(68mA) cut-off @RT - Discharge : 0.2C(680mA), 2.65V cut-off @RT *1C=3,400mA
3.2 Charging Voltage	4.2V
3.3 Nominal Voltage	3.60V
3.4 Charging Method	CC-CV (constant voltage with limited current)
3.5 Charging Current	Standard charge: 1,700mA For cycle life : 1,020mA
3.6 Charging Time	Standard charge: 4hours
3.7 Max. Charge Current	2,000mA (not for cycle life)
3.8 Max. Discharge Current	8,000mA (for continuous discharge) 13,000mA (not for continuous discharge)
3.9 Discharge Cut-off Voltage	2.65V
3.10 Cell Weight	50 g max
3.11 Cell Dimension	Height : Max. 65.25 mm Diameter: Max. Φ 18.55 mm
3.12 Operating Temperature (Cell Surface Temperature)	Charge : 0 to 45°C Discharge : -10 to 60°C
3.13 Storage Temperature	1 year : -20~25°C (1*) 3 months : -20~45°C (1*) 1 month : -20~60°C (1*)

Fig. 27: Samsung INR18650-35E nominal specifications (High Voltage)

Item	Specification
3.1 Standard discharge Capacity	Min 4,900mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 0.2C(980mA), 2.5V cutoff @ RT * 1C = 4,900mA
3.2 Rated discharge Capacity	Min 4,753mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 1C(4,900mA), 2.5V cutoff @ RT
3.3 Charging Voltage	4.2V
3.4 Nominal Voltage	3.6V
3.5 Charging Method	CC-CV (constant voltage with limited current)
3.6 Charging Current	Standard charge: 2,450mA
3.7 Charging Time	Standard charge: 3hours
3.8 Max. Charge Current	4,900mA (not for cycle life)
3.9 Max. Discharge Current	9,800mA (for continuous discharge) 14,700mA (not for continuous discharge)
3.10 Discharge Cut-off Voltage	2.5V
3.11 Cycle life	Capacity ≥ 3,802mAh @ after 500cycles (80% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, CCCV 0.05C(245mA) cut-off @ RT - Discharge: 1C(4,900mA), 2.5V cut-off @ RT
3.13 Recovery characteristics	Capacity recovery (after the storage) ≥ 4,278mAh (90% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Storage : 30 days @ 60℃ - Discharge : 1.0C(4,900mA) 2.50V cut-off @ RT
3.14 Cell Weight	69g max
3.15 Cell Dimension	Cell height : Max.70.80mm Diameter : Φ Max.20.25mm

Fig. 28: Samsung INR21700-50E nominal specifications (Low Voltage)

Low-voltage

The conception of the low-voltage battery is compulsory to power the electronic components responsible for the motor control as well as the collection of real-time data. Here is a non-exhaustive list of the components as well as nominal information on their energy needs.

Components	Quantity	Maximum DC Voltage (V)	Current (mA)
Arduino Every ABX00028	6	21V	250
Accelerometer MPU-6050	1	5V	3.9
U-Block NEO-6_M	1	3.6V	67
Raspberry Pi 4b	1	6V	3000
CAN translater MCP-2515	7	5.5V	35
Actuators	2	12V	10000
2.4GHz Antenna	1	24V	500
Thermistors	20	Less than 24	200
Current sensors	5	Less than 24	20
Wiring Relay	4	Less than 24	20
Brake Relay	1	Less than 24	5

Fig. 29: List of low-voltage components

As the low-voltage components are connected in parallel, we need to ensure a minimum voltage of 24V and a nominal discharge current of 14.1A. So, in order to size our low-voltage battery correctly, we will outline the requirements that our future battery should meet.

Autonomy: 30 minutes
Capacity: 19A
Nominal voltage: 24V

We intend to put before each component a voltage controller to adjust the voltage to its nominal requirements.
Tp size our low-voltage battery, we selected lithium cells that match the desired characteristics using the lithium cell database on the "kaispower" website. Subsequently, we compared them in terms of voltage, capacity, nominal discharge current, and price; testing various battery configurations.

Here is an overview of our research process:

COMPARAISON DES CELLULE

		Tension min (V	Tension (V)	Tension max (V	Capacité (Ah)	I continue (A)	I crête (A)	Prix / u
1	Samsung INR21700-48G	2,5	3,6	4,2	4,7	9,6		5,65€
2	Samsung INR21700-50E	2,5	3,63	4,2	4,9	9,8	14,7	4,70€
3	LG INR18650MJ1	2,5	3,635	4,2	3,4	10		3,30€
4	PanasonicNCR18650GA	2,5	3,6	4,2	3,35	10		3,15€

ESSAI BATTERIE 7s2p 14 cellules

Cellules	Tension min (V	Tension (V)	Tension max (V	Capacité (Ah)	I continue (A)	I crête (A)	Wh	Prix	Prix/Wh
Samsung INR21700-48G	17,5	25,2	29,4	9,4	19,2	0	236,88	79,10€	0,33€
Samsung INR21700-50E	17,5	25,41	29,4	9,8	19,6	0	249,018	65,80€	0,26€
Panasonic NCR18650GA	17,5	25,2	29,4	6,7	20	0	168,84	44,10€	0,26€

ESSAI BATTERIE 7s3p 21 cellules

Cellules	Tension min (V	Tension (V)	Tension max (V	Capacité (Ah)	I continue (A)	I crête (A)	Wh	Prix	Prix/Wh
Samsung INR21700-48G	17,5	25,2	29,4	14,1	28,8	0	355,32	118,65€	0,33€
Samsung INR21700-50E	17,5	25,41	29,4	14,7	29,4	44,1	373,527	98,70€	0,26€
Panasonic NCR18650GA	17,5	25,445	29,4	10,2	30	0	259,539	69,30€	0,27€

ESSAI BATTERIE 7s4p 28 cellules

Cellules	Tension min (V	Tension (V)	Tension max (V	Capacité (Ah)	I continue (A)	I crête (A)	Wh	Prix	Prix/Wh
Samsung INR21700-48G	17,5	25,2	29,4	18,8	38,4	0	473,76	158,20€	0,33€
Samsung INR21700-50E	17,5	25,41	29,4	19,6	39,2	58,8	498,036	131,60€	0,26€
Panasonic NCR18650GA	17,5	25,2	29,4	13,4	40	0	337,68	88,20€	0,26€

Fig. 30: Low-Voltage Battery Dimenioning Excel

Based on what has been explained above, we can see that our choice has been made for the "Samsung INR21700-50E" cell in a 7s2p configuration due to its very interesting characteristics and moderate cost.
Furthermore, this configuration allows us to use only 14 cells, wich will significantly contribute to reducing the weight and size of our battery.

Based on the characteristics of our low-voltage cell, we can calculate the theoretical autonomy, weight, and size of our battery :

Autonomy : Autonomy $= \frac{9.8}{14.1} = 0.70\,h = 42min$

Weight : According to the technical documentation of the "Samsung INR21700-50E" cell, it weights 69g each.
Weight $= 14 \cdot 69 = 966\,g$

Size : According to the technical documentation of the "Samsung INR21700-50E" cell, it measures 21mm in diameter and 70mm in height.
Length $= 7 \cdot 21\,mm = 14.7\,cm$
Width $= 2 \cdot 21\,mm = 4.2\,cm$
Height $= 70\,mm = 7\,cm$

Item	Specification
3.1 Standard discharge Capacity	Min 4,900mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 0.2C(980mA), 2.5V cutoff @ RT * 1C = 4,900mA
3.2 Rated discharge Capacity	Min 4,753mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 1C(4,900mA), 2.5V cutoff @ RT
3.3 Charging Voltage	4.2V
3.4 Nominal Voltage	3.6V
3.5 Charging Method	CC-CV (constant voltage with limited current)
3.6 Charging Current	Standard charge: 2,450mA
3.7 Charging Time	Standard charge: 3hours
3.8 Max. Charge Current	4,900mA (not for cycle life)
3.9 Max. Discharge Current	9,800mA (for continuous discharge) 14,700mA (not for continuous discharge)
3.10 Discharge Cut-off Voltage	2.5V
3.11 Cycle life	Capacity ≥ 3,802mAh @ after 500cycles (80% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, CCCV 0.05C(245mA) cut-off @ RT - Discharge: 1C(4,900mA), 2.5V cut-off @ RT
3.13 Recovery characteristics	Capacity recovery (after the storage) ≥ 4,278mAh (90% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Storage : 30 days @ 60℃ - Discharge : 1.0C(4,900mA) 2.50V cut-off @ RT
3.14 Cell Weight	69g max
3.15 Cell Dimension	Cell height : Max.70.80mm Diameter : Φ Max.20.25mm

Fig. 31: Samsung INR21700-50E nominal specifications

High Voltage

We have four 48V AmpFlow motors to power to propel the capsule. Therefore, we have chosen to independently power these four motors for the sake of simplifying the complete system. We have four high-voltage modules, each composed of a battery (13s2p) of 48 V and 16 A continuous and 26 A peak, a BMS, the three associated a fuse, a relay, an E-stop, a manual disconnect, a PWM, and a motor.

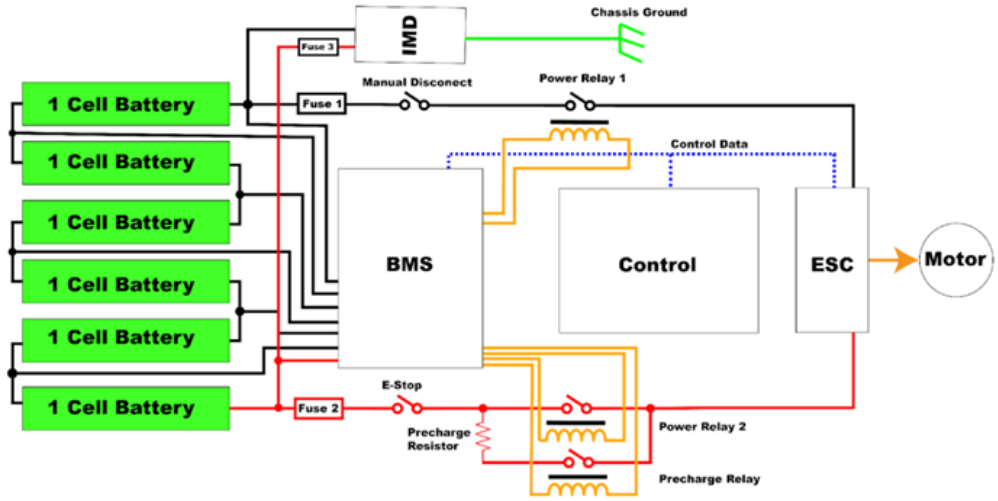


Fig. 32: High voltage schematic

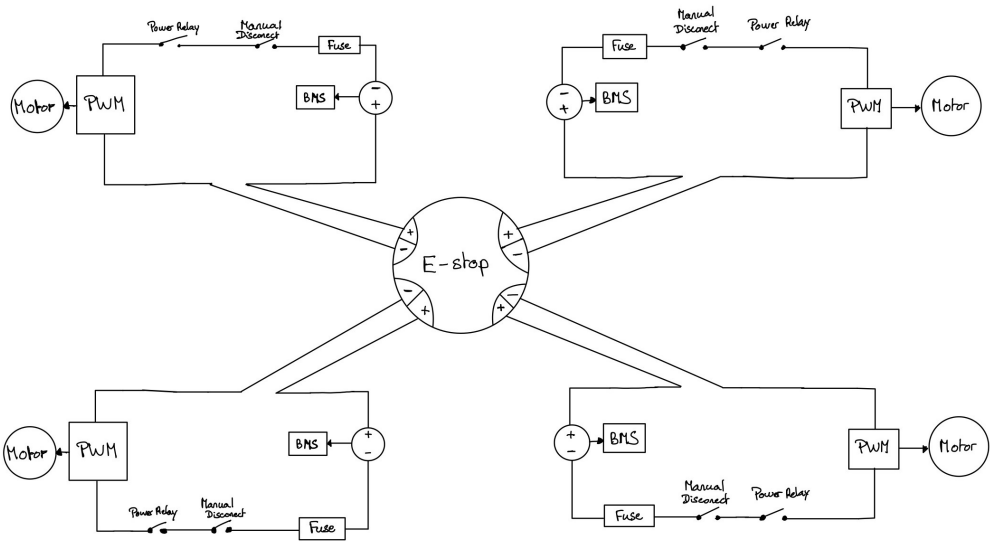


Fig. 33: High-Voltage Map

In order to operate the motor optimally, each high-voltage battery must provide a nominal voltage of 48V, a nominal discharge current of 15A, and a maximum discharge current of 25A.

So, in order to size our high-voltage battery correctly, we will outline the requirements that our future battery should meet.

Autonomy : 25 minutes
Nominal discharge current : 15A
Maximal discharge current : 25A
Nominal voltage : 48V

Like the low-voltage battery, to size our high-voltage battery, we selected lithium cells that match the desired characteristics using the lithium cell database "kaispower" website. Subsequently, we compared them in terms of voltage, capacity, nominal discharge current, and price; testing various battery configurations.

Here is an overview of our research process:

COMPARAISON DES CELLUL

		Tension min (V	Tension (V)	Tension max (	Capacité (Ah)	I continue (A)	I crête (A)	Prix / u
1	Samsung INR21700-48G	2,5	3,6	4,2	4,7	9,6		5,65€
2	Samsung INR21700-50E	2,5	3,6	4,2	4,9	9,8	14,7	4,70€
3	LG INR18650MJ1	2,5	3,635	4,2	3,4	10		3,30€
4	PanasonicNCR18650GA	2,5	3,6	4,2	3,35	10		3,15€

ESSAI BATTERIE 7s2p 14 cellules

Cellules	Tension min (V	Tension (V)	Tension max (	Capacité (Ah)	I continue (A)	I crête (A)	Wh	Prix	Prix/Wh
Samsung INR21700-48G	17,5	25,2	29,4	9,4	19,2	0	236,88	79,10€	0,33€
Samsung INR21700-50E	17,5	25,2	29,4	9,8	19,6	29,4	246,96	65,80€	0,27€
Panasonic NCR18650GA	17,5	25,2	29,4	6,7	20	0	168,84	44,10€	0,26€

ESSAI BATTERIE 7s3p 21 cellules

Cellules	Tension min (V	Tension (V)	Tension max (	Capacité (Ah)	I continue (A)	I crête (A)	Wh	Prix	Prix/Wh
Samsung INR21700-48G	17,5	25,2	29,4	14,1	28,8	0	355,32	118,65€	0,33€
Samsung INR21700-50E	17,5	25,2	29,4	14,7	29,4	44,1	370,44	98,70€	0,27€
Panasonic NCR18650GA	17,5	25,445	29,4	10,2	30	0	259,539	69,30€	0,27€

ESSAI BATTERIE 7s4p 28 cellules

Cellules	Tension min (V	Tension (V)	Tension max (	Capacité (Ah)	I continue (A)	I crête (A)	Wh	Prix	Prix/Wh
Samsung INR21700-48G	17,5	25,2	29,4	18,8	38,4	0	473,76	158,20€	0,33€
Samsung INR21700-50E	17,5	25,2	29,4	19,6	39,2	58,8	493,92	131,60€	0,27€
Panasonic NCR18650GA	17,5	25,2	29,4	13,4	40	0	337,68	88,20€	0,26€

Based on what has been explained above, we can see that our choice has been made for the "Samsung INR18650-35E" cell in 13s2p configuration due to its very interesting characteristics and moderate cost. Furthermore, this configuration allows us to use only 26 cells for each battery, which will significantly contribute to reducing the weight and size of our battery. Based on the characteristics of our high-voltage cell, we can calculate the

theoretical autonomy, weight, and size of our battery:

Autonomy : Autonomy $= \frac{6.5}{15} = 0.43h = 26min$

Weight : According to the technical documentation of the "Samsung INR18650-35E" cell, it weights 50g each
Weight $= 26 * 50 = 1300g$
Total High-Voltage Weight $= 4 * 1300 = 5200g$

Size : According to the technical documentation of the "Samsung INR18650-35E" cell, it measures 18.55mm in diameter and 65.25mm in height.
Length $= 13 * 18.55mm = 24.115cm$
Width $= 2 * 18.55mm = 3.71cm$
Height $= 65.25mm = 6.525cm$

Item	Specification
3.1 Standard discharge Capacity	Min 3,350mAh - Charge : 0.5C(1,700mA), 4.2V, 0.02C(68mA) cut-off @RT - Discharge : 0.2C(680mA), 2.65V cut-off @RT *1C=3,400mA
3.2 Charging Voltage	4.2V
3.3 Nominal Voltage	3.60V
3.4 Charging Method	CC-CV (constant voltage with limited current)
3.5 Charging Current	Standard charge: 1,700mA For cycle life : 1,020mA
3.6 Charging Time	Standard charge: 4hours
3.7 Max. Charge Current	2,000mA (not for cycle life)
3.8 Max. Discharge Current	8,000mA (for continuous discharge) 13,000mA (not for continuous discharge)
3.9 Discharge Cut-off Voltage	2.65V
3.10 Cell Weight	50 g max
3.11 Cell Dimension	Height : Max. 65.25 mm Diameter: Max. Φ 18.55 mm
3.12 Operating Temperature (Cell Surface Temperature)	Charge : 0 to 45°C Discharge : -10 to 60°C
3.13 Storage Temperature	1 year : -20~25°C (1*) 3 months : -20~45°C (1*) 1 month : -20~60°C (1*)

Fig. 34: Samsung INR18650-35E nominal specifications (High Voltage)

BMS

Regarding funding concerns and system simplicity, we decided not to use the Orion Jr.2 BMSs used last year. We have chosen to use simple 13S 25A BMSs for the high-voltage batteries and a simple 7S 20A BMS for the low voltage. Even though these BMSs are not considered smart, we will add sensors to them (see details in the Electronics, Controls, and Communication section) to be able to receive information.

Relays

For the relays, we will use standard 48V 40A relays.

E-stop and manual disconnect

For the E-stop, we have chosen a large red emergency button that will be pressed in case of emergency to shut down the system. It will be connected to all four high-voltage systems.

Regarding the manual disconnect, we have selected a manual disconnect that can be activated with a key. The system requires two different individuals who will be assigned this responsibility. These individuals will be the head of the Electrical team. They will need to check every aspect of the wiring system before starting the pod.

Fuse

These are a component of our high-voltage system. The fuses are present to protect the high-voltage system from excessive current that could damage its various components.

We have decided to size our fuses based on the PWM, indeed, it is the component of our system capable of taking the least amount of current possible as input. Indeed, it can only take up to 25A max with a voltage of 48V

Sizing of fuse :

Given the information previously mentioned, we need to have a maximum current of 25A flowing through our circuit. However, for the sizing of the fuse, we need to have a factor of about 1.25 difference between the desired value and the actual value.
Fuse value $= 25A * 1.25 = 31.25A$
Thus, we can use a fuse rated for 30A at 48V.

Wiring

We use multi-strand cables of a size of 12AWG, which can therefore support a current beyond our maximum of 25A and 48V maximum
The color code is as follows:
- Red for the positive low voltage
- Black for the negative low voltage
- Orange with red every 30cm for the positive high voltage
- Orange with black every 30cm for the negative high voltage

For an additional distinction between the low-voltage circuit and the high-voltage circuit, we use different connectors for the two. Indeed, for the low-voltage system, the connector used for the connection with the battery is an XT60 connector, while for the connection between the high-voltage system and the battery, we use an XT90 connector.

2.8 Electronics, Controls, and Communication

The electronics of the pod were designed to be easy to use and program. With the lack of real experience of the team. We opted for the easiest solution. In the figure 35.. We highlight the main sections of the pod design and we will detail them in the following sections. The choice of using, Arduino, esp32, and Raspberry Pi was made because most of the team was familiar with these types of systems. It helped a lot in the design process because we didn't have to focus on the elements. We spent this time on the design and testing of our models.

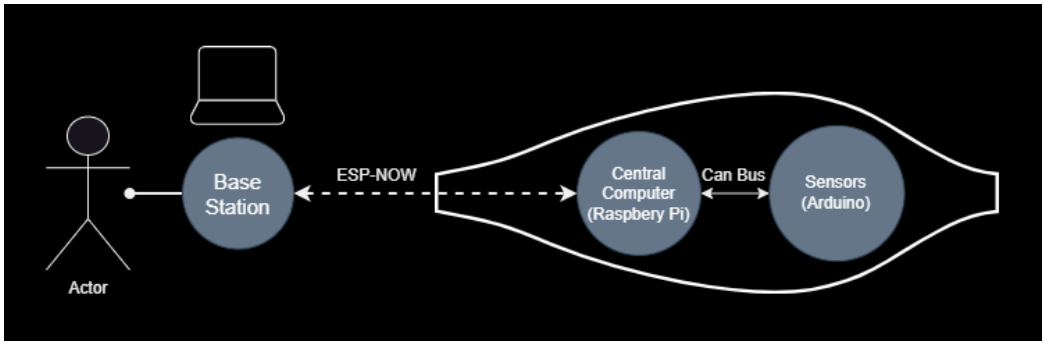


Fig. 35: Top level design

Base station

The base station is the user interface of the pod, allowing interaction with the pod and displaying data in real-time. It includes an emergency protocol in case of errors that have not been diagnosed by the central computer. Errors are prominently displayed, providing a clear understanding of the pod's status. Our initial design features a simple HTML page connected to an ESP32 receiver, displaying all key characteristics of the base station in real time (appendix A.5.1).

Antenna Communications

For the antenna communication, we opted for 2 esp32 linked via ESP-NOW. ESP-NOW is A wireless communication protocol designed by esp-32 to send in data in low-range applications (around 220 meters). We decided to use this software because it was easier to set up than many of the systems we studied. And it was also the most reliable. We already tested many messages sent across long distances and only around 180m we started to lose some packets. The track being 100m long we will be able to receive info from the systems at any time.

Despite the reliability, we had to fix an issue. We needed to connect the ESP receiver to the Raspberry Pi. And we decided to use the UART communication protocol to link the

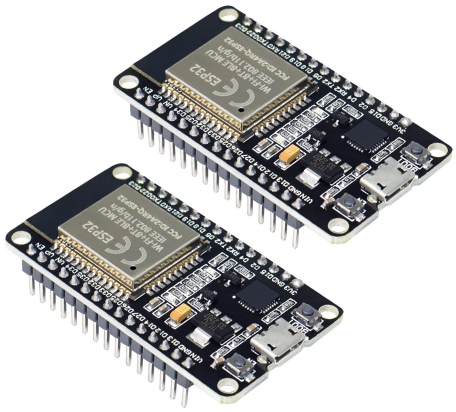


Fig. 36: ESP 32

two. It was by far the easiest and most reliable one. The data is sent to the UART by the pi in real time and the ESP receiver receives it and sends the new data to the Base station.

Onboard Computer

The onboard computing unit (Appendix A.5.2 and A.5.3) is the brain of the pod and is responsible for making critical decisions based on sensor data, coming from attached modules, and programmed states. The main computing unit consists of a Raspberry Pi4 running a fully custom program made in Python emulating a state machine. It uses various data structures and protocols to communicate to other modules (by CAN bus) or to the base station.

When we start to manually verify all the statistics of the pod we are making and all around check to verify that all the systems have been awakened and work properly. when everything has been verified we reset the variables that need to be sure to align the referential of the pod and we enter the launch state.

During the launching state, every safety check has been verified and the pod is waiting for our command to launch. When launched the pod accelerates, maintains, and breaks depending on the state. during these operations, we constantly do a safety check to verify the pod global variables. If anything happens the pod will enter an emergency mode (Battery related or not). After the run, we run a global safety check to be sure that everything works properly.

We decided to follow the LED display given by the rulebook. We only added a blue display for the default mode.

Can Communication

The CAN (Controller Area Network) communication protocol is a robust and widely used standard in the field of embedded systems and industrial applications. It is designed for high-reliability data transmission and is particularly well-suited for scenarios where multiple devices need to communicate efficiently and reliably on a shared bus.

In our prototype, we have implemented a CAN communication system. We have connected four Arduino boards, each connected with MCP2515 modules, and a Raspberry Pi 4 also equipped with a CAN hat module. The mcp2515 transforms the SPI communication of the micro-controllers into can messages. This setup forms the backbone of our communication infrastructure. Additionally, our prototype includes the integration of a Battery Management System (BMS). Through our research and development process, we have found that the CAN communication protocol offers the most reliable and efficient means of communication within our pod. It excels in providing a low-latency, high-throughput communication channel that ensures data integrity, making it an ideal choice for our application.

Electronics / Embedded Systems

Within the overall architecture of our capsule, we have identified several key sections, including the propulsion sections that share a similar design. Inside the capsule, we have integrated three major circuits (Fig. 37). Firstly, there is the positioning system, responsible for capturing data related to the capsule's position and acceleration. Next, we have the braking system, which manages the brake systems and includes infrared thermal sensors for monitoring temperature. Finally, there are the motor systems that control the capsule's propulsion. These three circuits are essential for the proper functioning of our device.

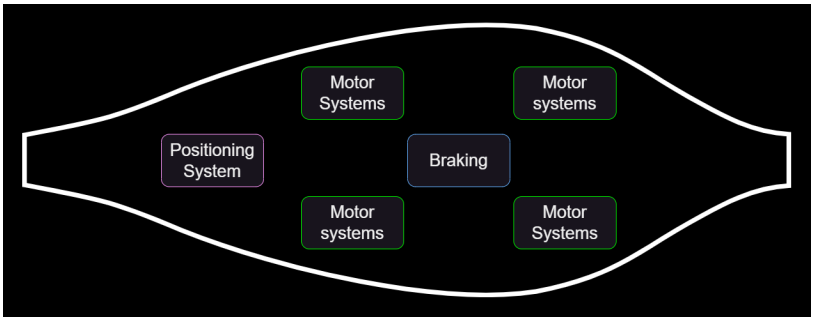


Fig. 37: Overview of the sensors sections

Positioning System:

- Utilizes GPS modules and accelerometers for determining position and acceleration.

Braking System:

- Control the actuators to activate the braking systems.
- Integrates infrared thermal sensors to measure the track temperature.

Motor Systems:

- Uses a motor controller for the management of propulsion systems.
- Incorporates thermal sensors, particularly thermistors, for temperature monitoring.
- Use a current sensor to estimate the current consumed by the high voltage system.
- Command the relays of the power systems.
- Read the command of a tachometer to estimate the distance.

It is important to note that each section includes a converter that transforms the 24V tension from the low-tension batteries into a 5V tension. This reduced tension is crucial for powering the sensors and the Arduinos controlling them. Indeed, each module is operated by an Arduino, which collects data from the sensors, performs basic calculations, and then transmits this information through the CAN BUS communication system.

In the following sections, we will detail each of the sensors and their specific role within the system.

Propulsion

To power the motors we use the Cytron MD25HV. It receives commands from the Arduino to power the motors. For a DC controller, it is quite easy to use and that is why we selected it.

Position & accelerometry

The positioning of the pod is a critical element that determines its behavior on the track. Accurate data acquisition and the robustness of our components are essential to ensure the proper functioning of our prototype. To achieve this goal, we have adopted a sensor fusion approach, combining an accelerometer and a GPS to determine the capsule's position, while utilizing optimization processes to enhance data accuracy.

GPS :

We selected the neom-9m for our GPS systems. We decided to use this one for its high refresh rate (1000Hz) and precision around 2.0m. We will be able to estimate our position

on the track easily. Considering a velocity around 5m/s we would be able at the maximum refresh rate to estimate the position every 0.005m. That is very accurate and with a lot of data, it is easier to average them to estimate a refined position.

GNSS		GPS+GLO+GAL+BDS	GPS+GLO+GAL	GPS+GLO	GPS+BDS	GPS+GAL
Sensitivity[5]	Tracking and nav.	-167 dBm	-167 dBm	-167 dBm	-166 dBm	-166 dBm
	Reacquisition	-160 dBm	-160 dBm	-160 dBm	-160 dBm	-160 dBm
	Cold start	-148 dBm	-148 dBm	-148 dBm	-148 dBm	-148 dBm
	Hot start	-159 dBm	-159 dBm	-159 dBm	-159 dBm	-159 dBm
Position accuracy	PVT	2.0 m CEP	2.0 m CEP	2.0 m CEP	2.0 m CEP	2.0 m CEP

Table 1: NEO-M9N-00B typical performance in multi-constellation GNSS modes

GNSS		GPS	GLONASS	BEIDOU	GALILEO
Acquisition[3]	Cold start	29 s	27 s	32 s	42 s
	Hot start	2 s	2 s	2 s	2 s
	Aided start[4]	2 s	2 s	2 s	5 s
Max navigation update rate	PVT	25 Hz	25 Hz	25 Hz	25 Hz
Sensitivity[5]	Tracking and nav.	-166 dBm	-166 dBm	-160 dBm	-159 dBm
	Reacquisition	-160 dBm	-156 dBm	-158 dBm	-154 dBm
	Cold start	-148 dBm	-145 dBm	-145 dBm	-140 dBm
	Hot start	-159 dBm	-156 dBm	-159 dBm	-154 dBm
Position accuracy	PVT	2.0 m CEP	4.0 m CEP	3.0 m CEP	3.0 m CEP

Fig. 38: Statistics of the NEOM-9m

Accelerometer:

We selected the MPU-6050 for its satisfactory refresh rate and high accuracy. This accelerometer allows us to collect data at high speed, translating into more precise measurements. With a refresh rate of 1,000 Hz and an error margin in the milli-G range, it effectively meets our requirements.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
ACCELEROMETER SENSITIVITY						
Full-Scale Range	AFS_SEL=0		±2		*g*	
	AFS_SEL=1		±4		*g*	
	AFS_SEL=2		±8		*g*	
	AFS_SEL=3		±16		*g*	
ADC Word Length	Output in two's complement format		16		bits	
Sensitivity Scale Factor	AFS_SEL=0		16,384		LSB/*g*	
	AFS_SEL=1		8,192		LSB/*g*	
	AFS_SEL=2		4,096		LSB/*g*	
	AFS_SEL=3		2,048		LSB/*g*	
Initial Calibration Tolerance			±3		%	
Sensitivity Change vs. Temperature	AFS_SEL=0, -40°C to +85°C		±0.02		%/°C	
Nonlinearity	Best Fit Straight Line		0.5		%	
Cross-Axis Sensitivity			±2		%	
ZERO-G OUTPUT						
Initial Calibration Tolerance	X and Y axes		±50		*mg*	1
	Z axis		±80		*mg*	
Zero-G Level Change vs. Temperature	X and Y axes, 0°C to +70°C		±35			
	Z axis, 0°C to +70°C		±60		*mg*	
SELF TEST RESPONSE						
Relative	Change from factory trim	-14		14	%	2
NOISE PERFORMANCE						
Power Spectral Density	@10Hz, AFS_SEL=0 & ODR=1kHz		400		µ*g*/ √Hz	
LOW PASS FILTER RESPONSE						
	Programmable Range	5		260	Hz	
OUTPUT DATA RATE						
	Programmable Range	4		1,000	Hz	
INTELLIGENCE FUNCTION INCREMENT			32		mg/LSB	

Fig. 39: Statistics of the accelerometer

Sensor Fusion:

To maximize the accuracy of our data, we collaborated with a professor specializing in optimization. The GPS has a 'slow' sampling to be accurate compared to the accelerometer, and the accelerometer can sample faster but it will develop natural errors over the multiple integrations to estimate the position and speed. To solve this issue we use sensor fusion. We merge the data of the 2 sensors. we merge them using the Uniformly-Accelerated Motion equation and computer integration. we can reduce the global error.

This sensor fusion approach ensures that our system has reliable and precise positioning data, which is crucial for the safety and overall performance of the capsule.

Temperature

To ensure proper heat control in the capsule and its components, we employ two heat detection systems. The first system consists of thermistors and the second uses infrared temperature sensors.

Thermistors

Thermistors offer a simple and effective method for monitoring the temperature of various elements both inside and outside the capsule. They are directly attached to areas of the capsule that generate the most heat. These thermistors play a crucial role in the thermal control of our systems, particularly the motors and the capsule's external surface. This approach allows us to monitor temperature fluctuations in real time and take appropriate measures accordingly.

Infrared thermal sensors

For our braking system, which incorporates a magnetic brake this year, we have chosen to use infrared temperature sensors. This decision stems from the fact that induction braking can lead to an increase in temperature in the rail. To validate our braking models and ensure they do not harm the rail, we decided to implement infrared thermal sensors.

We have selected the MLX90614 from Melexis for this task. With a measurement uncertainty ranging between ± 0.5 degrees Celsius and ± 2 degrees Celsius for estimated temperatures, this sensor provides sufficient accuracy for our needs. Furthermore, it integrates easily with an Arduino, facilitating the collection and transmission of data. It is also supported by numerous libraries that simplify data retrieval from the sensor's PWM signal.

However, we are aware of a major challenge: the emissivity of aluminum. Aluminum has low emissivity in the infrared spectrum, meaning it reflects less infrared radiation than usual. This property can introduce an error in our temperature measurements, as aluminum tends to absorb some of the infrared radiation. Currently, we are in the experimentation and research phase to find solutions aimed at minimizing this error. This is one of the major challenges we face in designing our system, and we are actively working to resolve this issue.

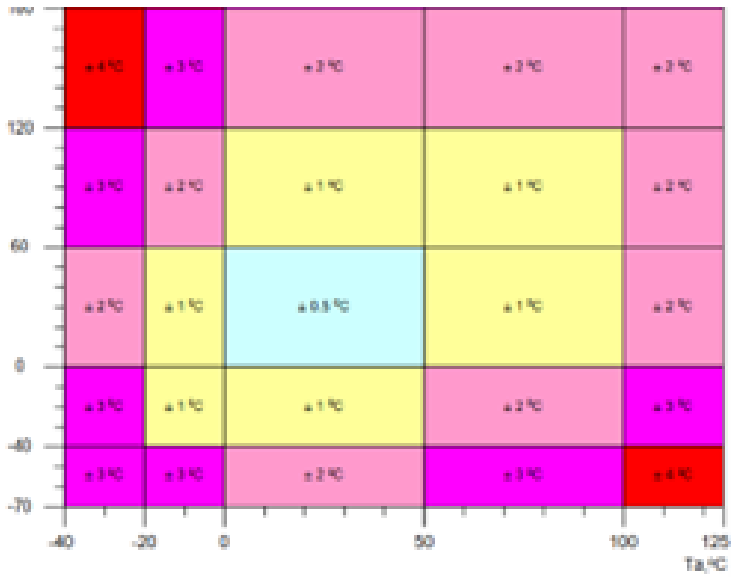


Fig. 40: Statistics of the infrared thermometer

Energy

Current sensor

To monitor consumption, we use current sensors at the outputs of the high-tension (HT) and low-tension batteries. This enables us to track the energy consumed by our systems. The objective is to be able to provide consumption data.

The BMS

The BMS is connected to the CAN communication systems and is equipped with a multitude of sensors whose information we utilize. These include:

- Battery charge percentage
- Battery temperature
- Charge of each cell
- ...

A detailed description of the BMS will be provided in the Power Supply section.

2.9 Pressurized Systems

The 2024 Polyloop hyperloop pod will not have any pressurized system. Therefore no overview of the design will be provided.

3 Subsystems description (Manufacturing)

3.1 Aeroshell

Manufacturing process

This year, aeroshell is made possible through a partnership with Fibrotek. Fibrotek Advanced Materials is known as a vertically integrated and innovative solution provider producing composite tooling and high-precision machined assemblies to the aerospace, nuclear, mining, and healthcare industries.

In exchange for some terms, they produce our unique carbon fiber shape aeroshell to our specifications and tolerances.

We provide Fibrotek with a polyurethane panel and they machine a master mold. Then, they laminate it to perform an infusion (Fig. 42). Finally, it goes in an autoclave at 50°C. The high pressure and heat inside the autoclave are essential for producing a result with the desired fiber content. Another reason manufacturers use autoclaves is their effectiveness in reducing residual porosity, also known as void content. Their autoclave (Fig. 41) is 9 feet in diameter and 24 feet long. It can apply pressure up to 225 PSI (about 15.3 atm) and has a max temperature of 288°C (550F).

Fig. 41: Fibrotek autoclave

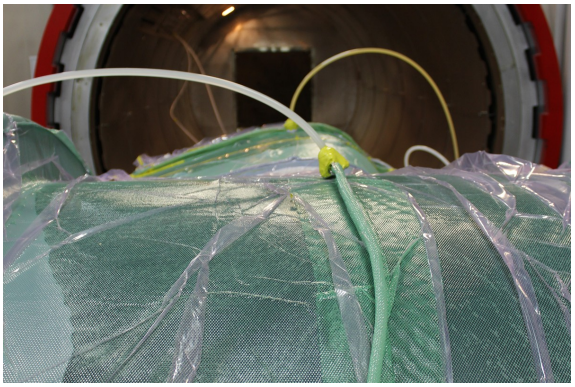


Fig. 42: Fibrotek infusion process

The master mold material is the Freeman M-3700 Modeling Board. Its specs are presented in appendix B.1.1. Freeman M-3700 offers a higher density and physical properties. Freeman M-3700 panel machines well and it is ideal for prototyping, as well as styling, designing and architectural models.

Once the aeroshell is built and delivered, a hood is cut out. As a matter of fact, the pod is made out of a monocoque, which ensures the fit. The use of a laser level is expected to draw the cutting line. Then, using a Dremel tool, the hood is cutout (Fig. 7). Those two parts are joined together with a magnetic sheet in a band via a thin carbon fiber sheet glued onto the inside edge used as a flange.

Lastly, the sponsor's logos are placed on the outer shell using white vinyl stickers on the dark carbon fiber shell. The contrast between the stickers and the shell is striking, accentuating the bold presence of our sponsors' logos against the sleek backdrop of the dark carbon fiber shell. This juxtaposition not only highlights the esteemed names and brands supporting our endeavors but also adds a dynamic visual element to our outer shell, embodying the spirit of innovation and collaboration that drives us forward.

Budget

The aeroshell is the image of Polyloop. It composes the front end on all our pictures. It represents us. Therefore, it is important to master the aeroshell. The team decided to invest in their picture. Most of the mechanical team budget is going toward the aeroshell. In fact, Fibrotek charges 1360$ for the raw materials (Epoxy resin and carbon fiber) and the infusion process. The part of the deal is that they cover the mastermold machining and do not charge the manual nor the infrastructure time. We add to acquire the mastermold via an other sponsor.

3.2 Structures/Chassis

Manufacturing process

As mentioned in section 2.2, the structure is made of 1 inch square 6061 T6 aluminum tube. The frame is designed to resist the vibration and different load combinations described in section 2.2 and shown in figure 43.

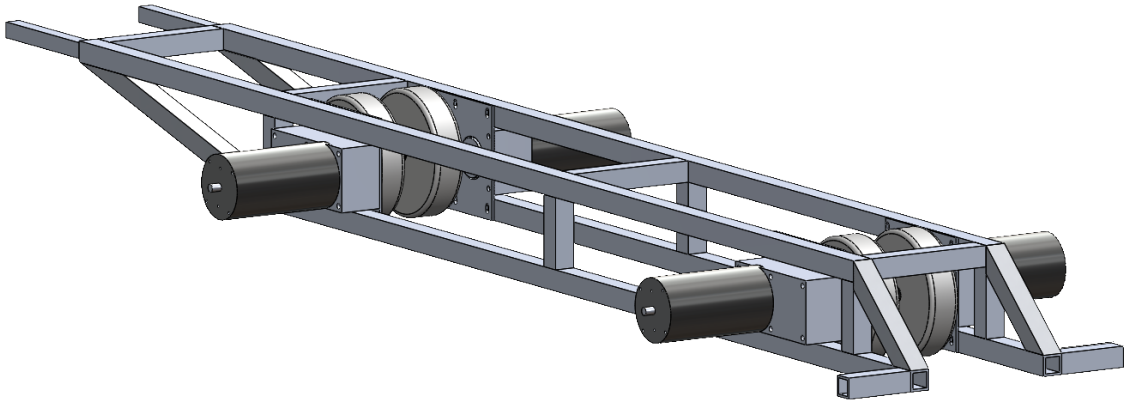


Fig. 43: Isometric view of the pod's structure

Even though using mechanical fasteners such as bolts and nuts offers lots of advantages over welding (Lower costs, Equal strength, Greater accuracy, Easier to finish, Easier to replace), we chose to weld the assembly.

Indeed, welding is a manual activity, so it carries a greater risk of human error that could result in costly rework. There's a common misconception that welding is stronger than bolting, but that simply isn't true. Bolting is just as strong as welding and can be stronger in some situations. The strength of a weld is largely determined by the expertise of the welder. The chances of an inexperienced welder applying a bad weld are much greater than the chances of a bolt failing. If you want your part to appear pristine after finishing operations, bolt-on is the way to go. Nothing lasts forever. Bolt-on parts are much easier in replacement scenarios because they can be done so precisely, whereas a weld has to be cut off in order to replace parts. It's also easier to reattach replacement parts onsite with bolt-on.

In our case, it's easier to achieve tight tolerances and more precision with welds. Also, we want the assembly to be permanent. This makes them a better choice for weight distribution. Also, we do not need to replace parts. We design the pod to perform in a single event.

Welding produces a permanent, physical connection between metals. Doing so, it delivers

high levels of strength and structural performance. Welding eliminates the need to perforate materials, as is the case with some methods of fastening. This characteristic can contribute to the structural integrity of welded installations. Welding is often the preferred method in many applications, due to contractor familiarity, work crew experience, tension and load characteristics, and other considerations. Welding aluminum presents several challenges, including:

- **Heat Dissipation:** Aluminum has high thermal conductivity, making it challenging to concentrate heat in a specific area during welding, which leads to warping, distortion, and burn-through.
- **Softness:** Aluminum is a relatively soft metal, making it prone to deformation during welding. It is crucial to use proper clamping and fixturing to prevent distortion.

We should definitively study the thermal impact of welds and make a weld map to ensure the integrity of the structure.

The chassis is welded using 4043 aluminum wire .035" in diameter through a Gas metal arc welding (GMAW) process. The choice of filler metal is based on figure 44 where the filler metal is suggested depending on the base metal. The advantages of 4043 filler are:

- Good wetting
- Less prone to cracking
- Better penetration
- Lower cost
- Less susceptible to black smut

We are equipped with a Millermatic 252 with a Miller Spoolmatic 15A air-cooled spool gun. The GMAW welding process has many advantages such as high-quality welds with good appearance and low spatter, better control over the welding process with less clean-up required, higher welding speeds and efficiency, good for welding thin materials, and easiest type of welding to learn. We've had GMAW aluminum welding course, also we will test our welds with a tension test to characterize the weld. Polytechnique Montreal has tension

machine[1] that go to 100 kN of force available to use. We will follow the ASTM E8/E8M - 22: Standard Test Methods for Tension Testing of Metallic Materials norm (*Standard Test Methods for Tension Testing of Metallic Materials*, 2022) to perform the tension test on our welds. The test will give an order of magnitude of the resistance our weld support. Also, we will characterize the penetration of the weld and adjust welding parameters in consequence.

Base Metal to Base Metal	1060, 1100, 3003, Alclad 3003	2219, A201.0	3004, Alclad 3004	5005, 5050	5052, 5652	5083, 5456	5086, 514.0, 535.0	5154, 5254	5454	6005, 6061 Alclad 6061, 6063, 6351	7005	354.0 C355.0	356.0, A356.0, 357.0, A357.0, 359.0, 443.0 A444.0
356.0, A356.0, 357.0, A357.0, 359.0, 443.0, A444.0	4043	4145	4043	4043	4043	5356	5356	4043	4043	4043	4043	4145	4043
354.0, C355.0	4145	4145	4145	4145	4043	NR	NR	NR	4043	4145	4145	4145	
7005	5356	4145	5356	5356	5356	5556	5356	5356	5356	5356	5356		
6005, 6061, Alclad 6061, 6063, 6351	4043	4145	5356	4043, 5356	5356, 4043	5356	5356	5356	5356	4043, 5356			
5454	4043	4043	5356	5356	5356	5356	5356	5356	5554				
5154, 5254	4043	NR	5356	5356	5356	5356	5356	5356					
5086, 514.0, 535.0	5356	NR	5356	5356	5356	5356	5356						
5083, 5456	5356	NR	5356	5356	5356	5556, 5183							
5052, 5653	4043	4043	4043	4043, 5356	5356								
5005, 5050	4043	4145	4043	4043, 5356									
3004, Alclad 3004	4043	4145	4043										
2219, A201.0	4145	2319											
1060, 1100, 3003, Alclad 3003	1100												

Fig. 44: Base metal to base metal chart

After the welding process, no heat treatment will be performed even though heat treating/annealing will be lost (maybe not totally) in the area where the weld is made. Our design safety factor is well over 2, thus ensuring the damage made on the heat treatment. For a structural component, a safety factor of 2 is not huge. There will be other things that degrade the safety factor down the line: quality of manufacturing, under and over-torqued bolts, and any misalignment in mounting or manufacturing. So for a prototype, a safety factor over 2 should be enough, but we may notice areas that are a little unstable or not performing as expected.

[1] https://sharedlab.bme.wisc.edu/labs/1002/equipment/mts/mts-insight.pdf

3.3 Guidance

Manufacturing process

Using the same system as our previous system, we only need to provide a mounting space on the aluminum structure. We also performed a visual inspection to ensure no previous apparent damages were present. We also replace the Belleville and rubber washers to ensure their performance.

Before the manufacturing process, measures of the internal structure are taken to consider the exact dimensions of the guidance system. After having mounted the caster on its foundation, we determine the number of rubber washers by trial and error to ensure a proper mechanical fit with the beam web.

3.4 Propulsion

Manufacturing process

The manufacturing process surrounding the propulsion is quite simple. We are improving last year's system. That being said, we are reusing the same motors and the same mounting system. As a result, we do not manufacture the front plate that holds the motors. Therefore, we only improve the last system by implementing Heavy Duty Unthreaded Bumpers (Fig. 45).

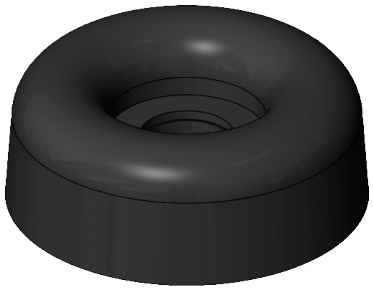


Fig. 45: Heavy Duty Unthreaded Bumpers

All of these bumpers are reinforced with an embedded washer or base plate for strength. Often used as feet, spacers, and vibration dampers, they absorb shock to prevent damage to equipment and surfaces. They're also good for starting and stopping motion on automated machines. We have to install them with screws. These bumpers are also known as recessed bumpers.

Bumpers with a hard and medium-hard rating withstand high levels of impact. The harder the bumper, the more impact it withstands. Bumpers with a medium hardness rating offer a balance of cushioning and impact resistance.

We use them as described in section 2.4 and circled in figure 46. They are installed with a screw through their center hole within the aluminum frame. They are carefully screwed with the right torque to avoid crushing the hollow aluminum tube.

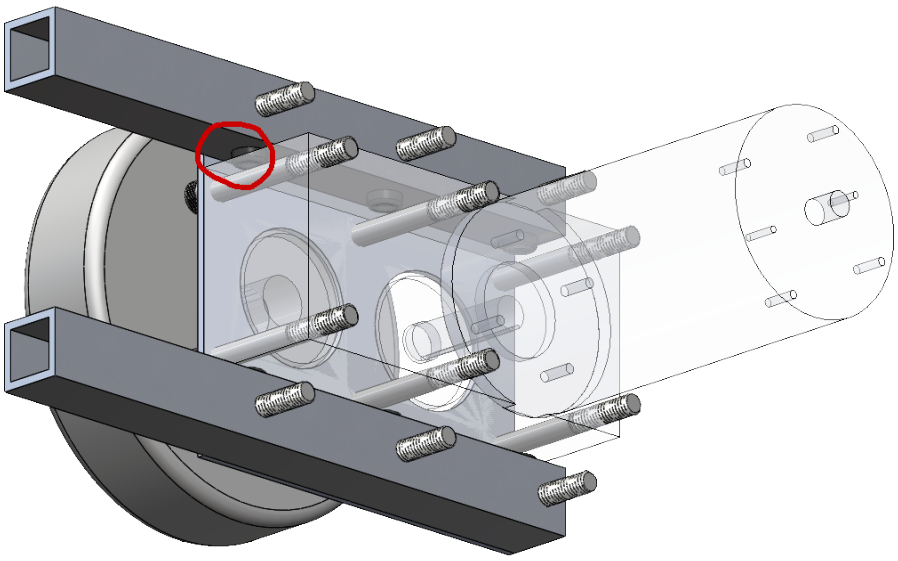


Fig. 46: Backside view of the propulsion system

3.5 Levitation

Manufacturing process

As mentioned in section 2.4, there won't be any levitating on this year Polyloop's prototype. The pod is using friction through contact with the rail to move forward. However, our physics team is working on a new LIM design, also presented in section 5.

3.6 Cooling Systems

Manufacturing process

As described in section 2.5, two students are working on their capstone project on the polyloop pod cooling system. Their goal is to do the conception of such cooling systems and not necessarily manufacture them.

The results of those projects will be useful for the next hyperloop pod to design an efficient cooling system as the electric power and the speed increase.

3.7 Braking

Manufacturing process

The braking system must not be made out of any magnetic metals, except the permanent magnet used, as it could prove highly difficult to assemble it. For that reason, it is mostly made of aluminum 6061 T6 as it is the one used by Polyloop. Most of the parts making up the braking system are welded together. This ensures a firm and sturdy fixture for the whole subsystem. The other parts are held together with aluminum bolts and screws that will go into the holes in fig. 47.

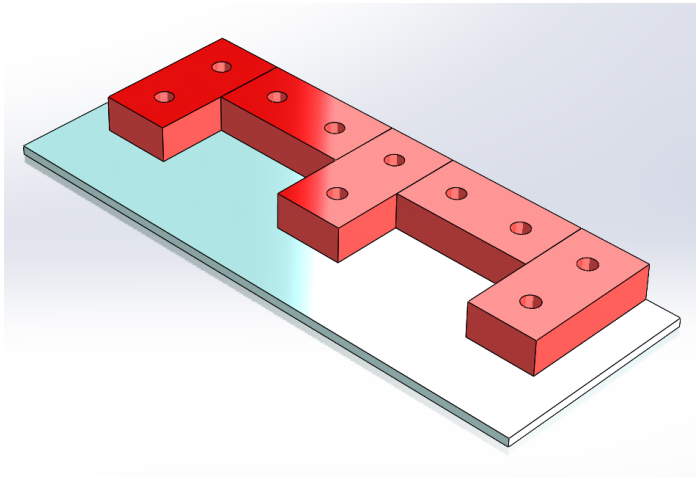


Fig. 47: Design of the Hallback array arrangement without the top plate.

The prototype is entirely manufactured by the student in charge of the project at the location at Polytechnique Montréal's technical companies workshop. The manufacturing stages are listed and presented below.

1. All hollow square aluminum extrusions and aluminum plates are cut to size by hand using a metal band saw
2. The holes in the screw/nut-fastened parts are drilled using a drill press
3. Components not fastened by screws/nuts are welded using a MIG welding post and an aluminum spool gun
4. The permanent magnets were carefully fixed in the predetermined positions, then secured with special countersunk aluminum screws
5. The rest of the components were assembled using screws/nuts/washers

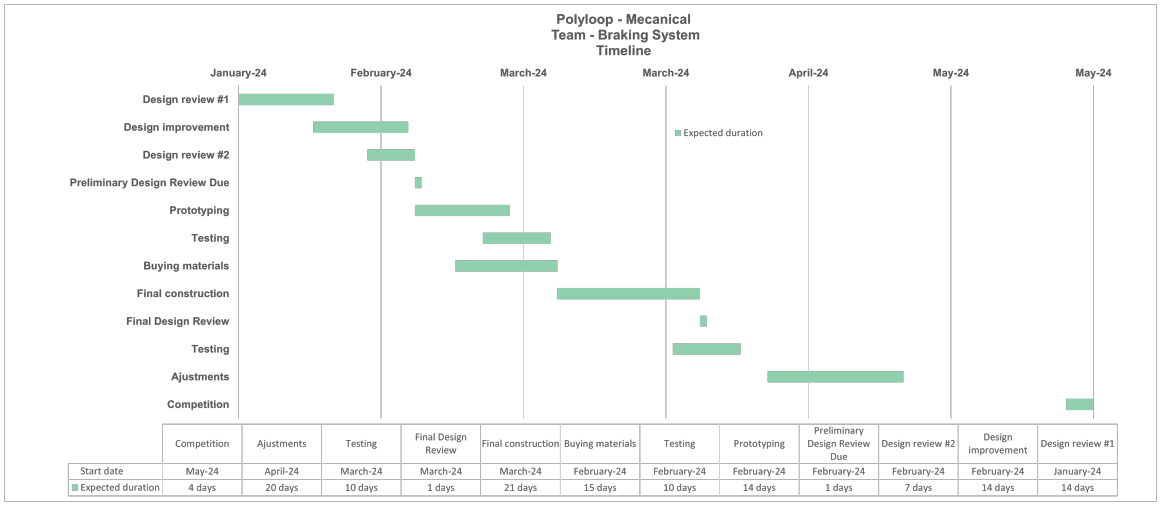
Safety and recommendations

The N52-grade neodymium magnets used in this assembly are extremely powerful, and handling must be handled with care to avoid any risk of injury or material damage. The following list presents the risks incurred by the user in the event of improper use.

- The power of these magnets can often surprise those unfamiliar with their strength. Fingers and other parts of the body can be crushed by the force of attraction produced between two magnets. The person working with them should wear thick gloves while handling them.
- These magnets are not intended for children; they are not toys.
- The strong magnetic fields of a neodymium magnet can affect pacemakers, ICDs, and other implanted medical devices. It is recommended that people with such devices stay away from these magnets.
- Neodymium magnets are made of a hard, brittle metal alloy, so they can come loose, chip, crack, or break if they collide. Protective eyewear must be worn when handling them, since in the event of breakage, small pieces can be thrown off at high speed.
- The strong magnetic fields produced can damage the magnetic media on floppy disks, credit cards, electronic cards, smartphones, computers, and CRT screens. We recommend neodymium magnets should not be kept near electronic devices.
- The maximum standard operating temperature for neodymium magnets is 80°C (175°F). Above this value, their operation is impaired and their service life is reduced.
- We strongly advise against drilling or machining neodymium magnets. When crushed to dust or powder, this material is highly flammable.
- The three-layer nickel-copper-nickel coating on these magnets presents a medical hazard for people allergic to this material. Such people should avoid prolonged contact with the product.
- IATA (International Air Transport Association) and US rules and regulations cover the shipment of magnets by air and ground. Magnetic fields can influence compasses or magnetometers used in air transport. They can also interfere with the internal compass of smartphones and GPS devices.
- Neodymium magnets can rust or corrode in the presence of moisture. If used underwater, outdoors, or in a damp environment, they can corrode and lose their magnetic strength.

Preliminary timeline

Figure 48 shows the manufacturing timeline of the braking system. Every step of the way is planned out. About 7 to 21 days is given for every step, depending on the estimation of time the task should take. Polyloop planned to have everything completed at least a month before the competition.



	Competition	Ajustments	Testing	Final Design Review	Final construction	Buying materials	Testing	Prototyping	Preliminary Design Review Due	Design review #2	Design improvement	Design review #1
Start date	May-24	April-24	March-24	March-24	March-24	February-24	February-24	February-24	February-24	February-24	February-24	January-24
Expected duration	4 days	20 days	10 days	1 days	21 days	15 days	10 days	14 days	1 days	7 days	14 days	14 days

Fig. 48: Timeline of manufacturing of the magnetic braking system

3.8 Power Systems

Manufacturing process

Busbar

The project team acquired bus bars from our sponsor, SIGMACLAD, which were then tailored to align with a predefined pattern, ensuring an optimal fit within the designated locations for our battery cells. The cutting of the bus bars was meticulously executed according to the specified pattern, as illustrated in the following fig. (49)

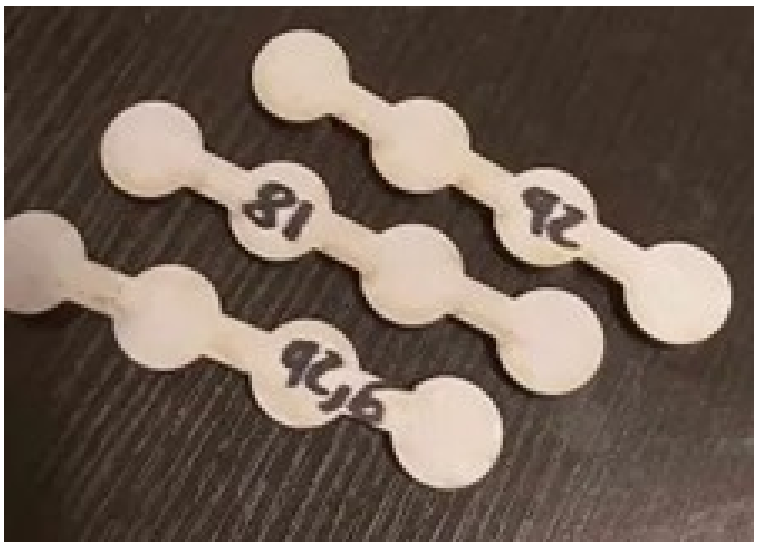


Fig. 49: Example of Bussbar with wood.

The precision cutting of the bus bars was facilitated through the utilization of laser cutting, following the established patterns. This method allowed for high accuracy and efficiency in preparing the bus bars for their intended application within the battery assembly process.

Battery

The decision was made to internally manufacture batteries, utilizing the Omega PA250i spot welding machine. This machine, acquired through university funding provided three years ago to support technical clubs at Polytechnique Montreal, is recognized for its industrial application, efficiency, and fast operational capability. The selection of this machine was guided by its availability and the benefits it offers, although it requires specific training for its effective use.

To streamline the manufacturing process of the batteries, spacers were employed. Furthermore, the assembly of the cells was facilitated using SIGMACLAD bus bars (Partnership). These bus bars, composed of nickel, steel, and copper alloy, were precision-cut precisely-cut? to meet the requirements of our battery construction.

Wiring

The wiring system began with a detailed mapping of the power system to ascertain the connectivity and configuration of components. This system incorporates four high-voltage batteries to energize four DC motors, alongside a single low-voltage battery dedicated to system operations.

Following the high-voltage system mapping, connections between components will be established and powered using an external source. This methodology is crucial for assessing the system's functionality before its integration into the capsule.

Battery casing

The final step involves the construction of a battery case, designed to safeguard the batteries against mechanical stresses, dust, and exposure. The construction of this case will be undertaken in-house, ensuring it is tailored to house the four high-voltage batteries, the low-voltage battery, and their respective Battery Management Systems (BMS). This approach not only provides protection but also aligns with the project/competition safety specifications.

Preliminary timeline

Regarding the project timeline, we are currently in the process of cutting the bus bars. During the March break, we will proceed with the soldering of the battery packs. At the same time, a team is dedicated to the wiring aspects of the capsule. By the end of the break week, our objective is to have completed the battery assembly and to strive for the operational readiness of at least one wiring system.

Following the assembly of the batteries, we will start the construction of the battery case, which will house all the capsule's batteries. Simultaneously, we will undertake testing phases.

Budget

The financial assessment for the project's battery components reveals that the low-tension (low voltage) battery will incur an approximate cost of $90. In contrast, the aggregate expense for the four high-tension (high voltage) batteries is estimated to be around $500.

Regarding the construction of the battery case, the budget is projected to be in the vicinity of $50. Lastly, the expenditure for the wiring is anticipated not to exceed $50, attributed to the availability of all necessary components already in possession for the capsule assembly. This budgetary allocation underscores the project's cost-management strategies, ensuring that essential components are procured within the designated financial parameters.

3.9 Electronics, Controls, and Communication

Manufacturing process

For the electronics, control systems, and communications, our project strategy does not consider in-house equipment manufacturing. Instead, our approach focuses on designing a highly reliable system by integrating pre-manufactured components. We choose this approach to prioritize the safety and reliability of the systems. The extent of manufacturing within our project is limited to the specific task of soldering components onto many printed circuit boards (PCBs) that we design. The PCB serves as the foundation for housing our electronic components, underscoring our commitment to precision in the design phase to ensure optimal system performance.

Preliminary timeline

The project timeline has undergone modifications due to alterations in team organization. Despite these changes, it remains imperative to achieve the completion of all subsystems by March 22$^{\text{nd}}$. This deadline was chosen to ensure we would have the time to assemble everything by the competition. The testing phase is scheduled for March 4$^{\text{th}}$. This phase is designed to test and review our electronics working together during the designated vacancy week. These objectives are critical to confirming the functionality of our design before the session concludes. Subsequently, the focus will shift predominantly towards the design and optimization of the PCB. This strategic approach ensures that despite many adjustments, the project remains on track to meet its critical milestones.

Budget

Regarding the budgetary considerations, a decision was made to re-purpose the majority of the equipment from the previous year. Consequently, the financial requirements for the current year's budget have been significantly reduced. The primary investments have been identified as the acquisition of a GPS sensor and the manufacturing of the printed circuit board (PCB).

3.10 Pressurized Systems

Manufacturing process

The pod doesn't have any pressurized system, thus no manufacturing process description has been done.

4 Subsystems description (Analysis and Testing process)

4.1 Aeroshell

The Teams did not conduct a wind tunnel experiment as expected. The amount of preparation behind this does not compare to the result we would have obtained.

The aeroshell is **not** under a lot of load. For the maximum reachable speed, we won't feel much drag nor lift. The aeroshell is mainly used to protect the interior components from wind, dust, and most importantly the sun.

A capstone project on the aeroshell revealed that the carbon fiber absorbs most of the heat. Having a light coating on top of the shell would reduce the heat significantly.

4.2 Structures/Chassis

With Polytechnque Montreal technician, we performed traction test on aluminium coupon to characterized the welds. The structure is built with hollow 1 inch square with 1/8" of thickness.

Polytechnique Montreal has tension machine[2] that go to 100 kN of force available to use. We followed the ASTM E8/E8M - 22: Standard Test Methods for Tension Testing of Metallic Materials norm (*Standard Test Methods for Tension Testing of Metallic Materials*, 2022) to perform the tension test on our welds. The test gave an order of magnitude of the resistance our weld support. On top of that, we characterized the weld penetration.

Those to results allowed us to adjust the welding parameters and assure a safety factor over 2.

[2] https://sharedlab.bme.wisc.edu/labs/1002/equipment/mts/mts-insight.pdf

4.3 Guidance

The guidance system consists of four fixed wheels positioned on both ends of the frame and on either side of the track. After considering the use of a spring-loaded caster in our initial design, we decided to use a fixed caster wheel. The spring-loaded caster wheels are usually made for heavier applications, thus bigger diameter wheels. Also, the springs that are used in such systems are too stiff for our needs.

Considering the mass of the fixed wheel is approximately $0,1$ kg, the maximum misalignment of the rail is 10 mm and the maximum speed of the capsule is 30 km/h, the maximum force of impact on the caster is around $100\,N$. Afterward, to realign the capsule, each caster will support $147\,N$ giving a security factor of around $2,6$. A system of rubber washers and Belleville washers is used to diminish the impact of the misalignment on the mechanical and electrical systems of the pods.

That lateral guidance system is tested on a 21 feet rail by pushing the aluminum structure by hand adjusting the number of rubber washers to satisfy a proper mechanical fit with the beam web.

As for the vertical guidance, we only test the stability, whether or not the propulsion wheel touches the rail simultaneously. Otherwise, final touches were made. It is important to ensure contact with the rail to maximize the propulsion.

4.4 Propulsion

As they were tested last year, we know the wheels motor work. We still perform no load test because we are prohibited to test inside the shop. And before we find an outdoor field to build our own track, we won't be able to test the entire pod.

However, we will test the whole system on the previous day of the competition, allowing us to make some minor changes if needed.

4.5 Braking

4.5.1 Failsafe system

The system was conceived to be a failsafe. There is a solenoid piston holding the array in a downward position. When we want to brake, we simply cut the power to the piston and it retracts itself, allowing the springs to bring up the array. The tests will consist of sending power to the piston and then shutting it down to assess the effectiveness of the system. We must ensure it works as intended; otherwise, it won't be failsafe.

4.5.2 Braking force

Simulations are a good starting point for forming hypotheses. However, we cannot solely rely on them to provide a complete representation of reality. This is why we must first experimentally verify that our magnetic brakes function as intended. We will gradually move them closer to a rotating disc and observe if it slow down. If it does, they are effective. To test them in the real-world scenario they will be used in, we will attach them to the internal structure and evaluate their performance at low speeds on a test track, gradually increasing the speed. The braking force should increase as the speed rises.

4.5.3 Magnetic shield

Permanent magnets cannot be turned off; their magnetic fields are always active. This means they are likely to attract magnetic objects that are in proximity to them. Additionally, being arranged in a Halbach array configuration extends their magnetic reach. To mitigate this effect, we must block or restrict the magnetic fields. One approach is to place a magnetic substance near the magnets, causing the magnetic fields to pass through the substance rather than through the air. There are various options for the shield material, such as iron, nickel, or steel. To test its effectiveness, we will experiment with different substances and geometries and observe how much the magnetic reach is reduced.

4.6 Power Systems

Security test

The BMS, fuses, and cable sizing greatly contribute to the safety system of our pod. Indeed, the power system team has spent time researching and sizing these components to ensure the capsule's optimal safety. To verify if the capsule goes into a safe state when a failure occurs within the capsule, we will be conducting tests over the next few weeks on our operating system to see how it reacts when a problem occurs.

Correct working of the manual E-stop test

Verifying the proper functioning of the manual E-stop is very simple and has already been started by our team. Indeed, we are currently in the process of wiring the hyperloop capsule, and each time a module is installed, we check its operation. Thus, to verify the manual E-stop, we start our system and then press the E-stop button to see if the system stops.

Heating test

The system heating check is continuously verified by our team, indeed, we are currently wiring the capsule. So far, under normal operating conditions for 10 minutes, the system has not heated up. We will continue the tests, to check the heat emitted as we approach the limit conditions of use.

Data test

When all systems are operational and the capsule is fully wired, we will conduct vacuum tests of the capsule to see how all the systems function together

4.7 Electronics, Controls, and Communication

For the Testing, the rulebook asked our section to provide a simulation of the pod going into emergency mode if something unexpected occurs. That's what we did. We made two videos showing the correct execution of the emergency. In the first one, the case station works according to the plan and we are simulating the movement of the pod. In the second one, we change during the simulation we change the value of the temperature sensor signaling that the pod temperature is around 60 degrees Celsius. And you can observe that the pod is running in emergency mode. We uploaded the test on YouTube to show the emergency. (Test Electronics)

5 Linear Electromagnetical Propulsion Project

The project objectives aim to design, manufacture, and successfully operate a linear induction motor. This motor is scheduled to be implemented in Polyloop's 2025 hyperloop pod as a first step towards frictionless propulsion. An examination of the initial prototype highlights identified flaws and the associated challenges. Consequently, the resulting design prioritizes function over performance. The construction of the electrical machine is entrusted to Delom Services, a local electric motor shop. The team presents a demonstration platform, providing a glimpse of the asynchronous motor's capabilities at reduced electrical power levels on the competition rail.

5.1 Learnings and insights

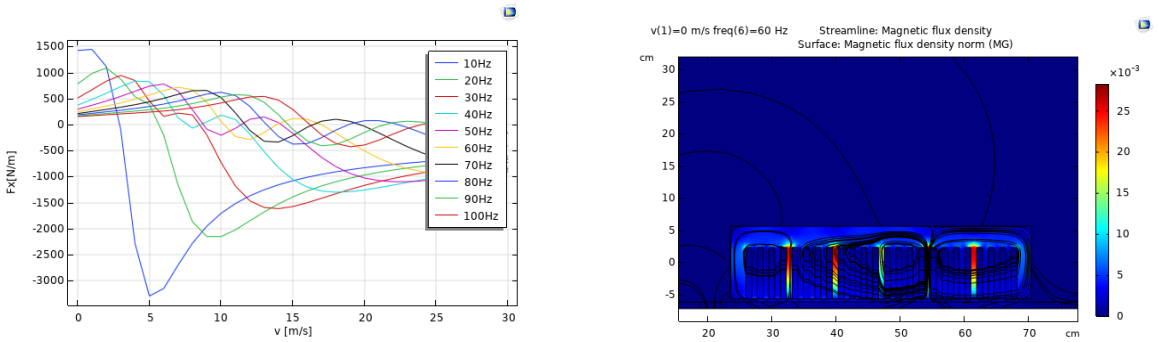
The initial prototype is the center of the case study for the present Physics team. Unfortunately, limited documentation is available for the device, as its development traces back to a prior R&D team. By the time it was powered for the first time, one of the motor's phases had received critical damaged. Nonetheless, its core was modified to build a simple 3-toothed single-sided linear induction motor, and the insight it provided was used in the design and manufacturing of the next prototype.

5.1.1 Old Model

Fig. 50: Initial prototype

To complete the development of a fully-fledged hyperloop pod, the propulsion system must incorporate a Linear Electrical Machine (LEM) utilizing the Lorentz force phenomenon for levitation. The initial design was finalized by early 2022, and the first motor, inspired by Timperio's master's thesis (Timperio, 2018), is a linear induction motor that was presented at the 2022 Canadian Hyperloop Competition. In the rush to showcase the engine at the event,

a crucial fabrication step was postponed with the intention to finalize it in the following months. However, by the beginning of the next semester, the team responsible for the motor design had departed from Polyloop, transferring the project to the electrical team. Lacking information about the engine and lacking the expertise to operate it, the project was ultimately set aside. In an effort to have a functioning capsule, the decision was made to implement DC motors. Thus, Polyloop's first LEM was shelved while information about it was slowly being recovered. Months passed, and the project was transferred to a newly formed physics team, who initiated the research. By the time the motor was powered for the first time, readings on the oscilloscope revealed that one of the phase groups had sustained fatal damage. Hence, the engine was inoperable. The cause of the damaged coil group was the lack of the final step in electric motor fabrication, the vacuum impregnation treatment. The process consist on solidifying the magnetic core wit a thick resin coat. Without such treatment, the motor's windings are sensitive to shocks and vibrations. The storage of the linear electrical machine was lacking, and proved to be its demise. Nonetheless, the research and preparation necessary for it's operation served as invaluable insights on how to design, conveice and operate a linear induction motor.



(a) Thrust Force at 200[A]

(b) Magnetic flux density

Fig. 51: Simulation of the first linear induction motor designed by Polyloop

Built out of laminations of electrical iron, the first prototype bore an impressive weight of 54[kg] and occupied a substantial volume, primarily attributed to the size of the coils as seen on figure 50. The prototype is notable for its size, yet details about its capabilities and operational power were unknown at the beginning of the project. To deduce its power output and operating specifications, the Physics team split into two groups. One focused on simulating the machines power, while the second concentrated on its actual operation. Utilizing COMSOL Multiphysics, the team recreated a 2D model of the electric motor to evaluate the theoretical thrust it could generate. Figure 51 illustrates the simulated engine's

force generation at 250[A] in a double-sided configuration. The engine was designed with a unilateral motor step-up, employing back irons to close the magnetic flux loops. However, as this marked the teams first venture into electromagnetical simulations, the double-sided configuration proved less complex to model due to the perfect magnetic conductor condition, which accelerated the simulation's convergence.

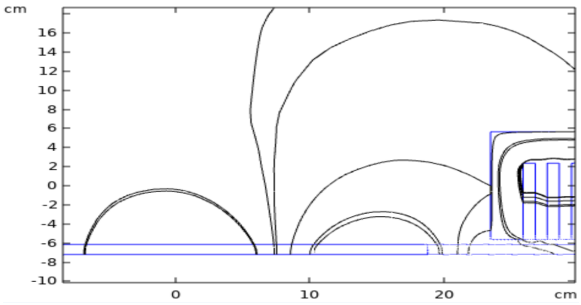


Fig. 52: Perfect magnetic conductor boundary's impact on flux lines

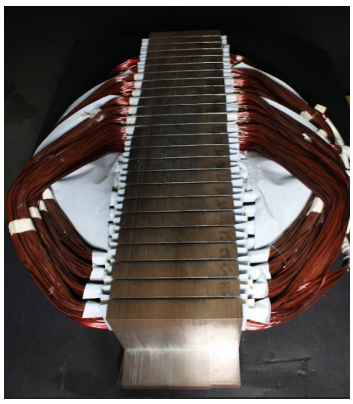
The simulation integrates the current density over the 2D modelled induction motor, providing the force per unit depth, as depicted in Figure 51a. Consequently, with a width of 113[mm], the resulting maximum thrust generated is 170[N] in a stationary state. As previously mentioned, this derived value represents two mirrored engines at the lower boundary, utilizing the perfect magnetic conductor boundary condition. This condition enforces the tangential magnetic component to be 0 at the surface interface. The result depicted in figure 52, where the flux lines are perfectly perpendicular at the lower bounds, representing a mirrored model. Thus, the expected output force is expected to be reduced by at least a factor of 2. Although imprecise, this data gave an upper bound utilized to prepare a low current test with proper safety guidelines. Thus, the faulty coils where detected.

While generating a theoretical Lorentz force of 85[N] , the power requirements of the electric motor surpass the current capabilities of the electrical team. At lower power levels, the device's mass hinders effective propulsion, and its geometrical specifications pose an unexpected challenge in the integrating the electrical machine.

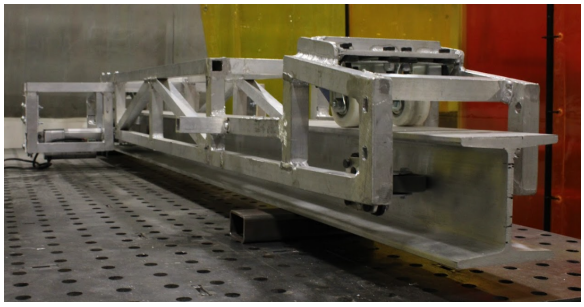
5.1.2 Integration Challenges

With a core composing of 24 teeth, the magnetic core utilized a three-phased distributed coil configuration. 12 turns of an equivalent 000 AWG multi-stranded wire composed each coil of the three phase groups. Therefore 200[A] is considered as the intended maximum

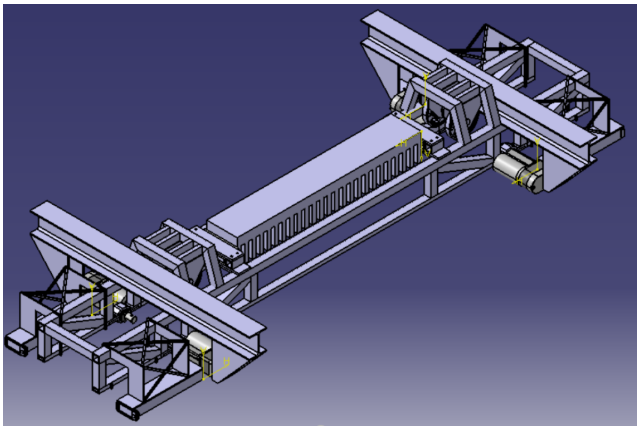
current intensity. The massive arrays of coils gave the entire electrical machine a 0.75 width to length ratio. Unfortunately, the magnetic flux density loss could not be quantified as the engine is no longer operational. Additionally, The first pod's geometry did not foresee space for the coil groups as portrayed in figure 53. The asynchronous machine occupies a



(a) Top view of the first prototype



(b) Internal structure of Clark 1



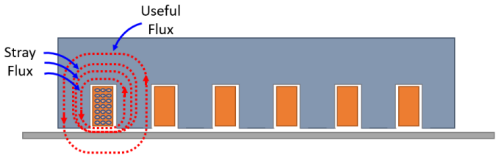
(c) Planned geometry for Clark 1

Fig. 53: Comparative perspectives of the first prototype's pod integration

significant mass and volume, presenting challenges, particularly concerning the precision required for optimal operation. The thrust derivation of figure 51a was simulated at a $3[mm]$ distance between the stator and rotor. It is essential to note that larger gaps result in increased flux losses. A recent discovery revealed that an overlooked aspect of the electric engine was the spacing between the teeth at the stator's lower boundary. During the design of the next iteration of this machine, the simulation group in the physics team observed that the spacing between the teeth should be greater than the space between the rotor and the stator. The first prototype has a spacing of $1[mm]$, resulting in the redirection of magnetic

flux within the stator itself. Figure 54b shows the flux losses due to field lines looping back before reaching the aluminum rail. The close spacing in between teeth offer the flux a lower reluctance path than the air between the rotor and stator, causing some flux lines not to penetrate the aluminum surface if the distance to the rotor is greater than the teeth spacing.

(a) Core Computer Model Close up



(b) Stray Flux Visualisation, Reproduced from (Timperio, 2018)

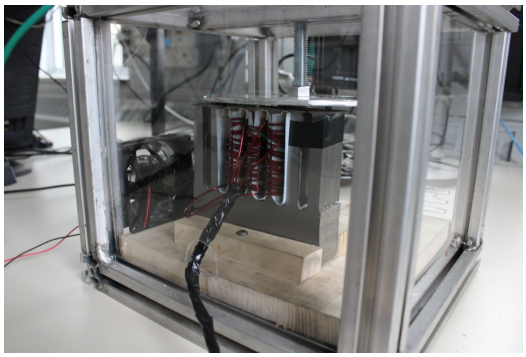
Fig. 54: Reluctance effects

5.2 Design

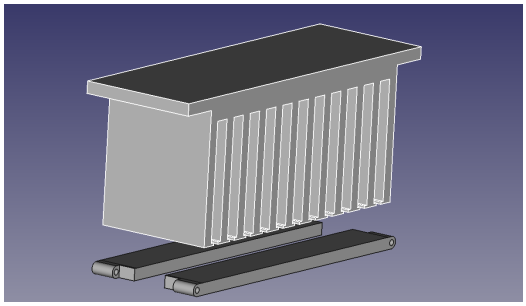
The first failed prototype led Polyloop to take a step back of the Lorentz force propulsion ambitions. A new prototype, shown in figure 55a was crafted using the recycled core of the initial engine, as the project budget was depended on the quality of deliverables. Constructed entirely from recycled materials, the first operational linear induction motor validated the team's simulation process. Consequently, the development for the next iteration of the asynchronous motor is authorized, and the resulting design is illustrated in Figure 55b. The mandate specifies the delivery of a linear electrical machine capable of accelerating a light pod to a speed of $5[m/s]$. Constraints and guidelines from Polyloop's upper administration require the prototype to be functional but not powerful, minimizing the effort needed for its integration. Thus, the Physics team focused on designing a linear induction motor and optimizing thrust generation at a lower current of $50[A]$ while tailoring the geometry to permit operation at $120[A]$.

5.2.1 Windings

The research concluded that concentric windings, while not as efficient as distributed windings, lead to a lighter and leaner design. Leading the development towards a simplified motor. Budget constraints encouraged such configuration, as the total length is reduced when a coil winding spans no further than one tooth. The development of next prototype



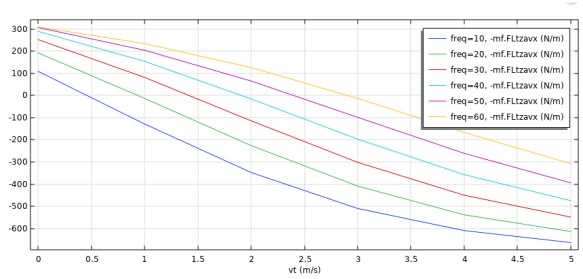
(a) Experimental Setup of a small Induction Motor



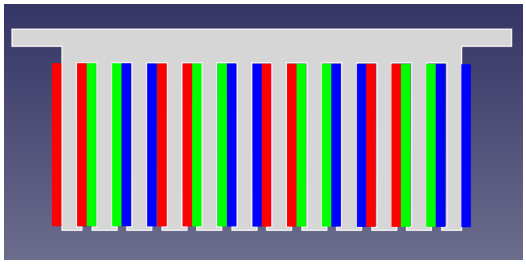
(b) Computer Assisted Design of the Next Prototype

Fig. 55: Electrical Machines of Polyloop

commenced by a simulation effort. An analysis of the force generated by different coils per phase group revealed a quasi-linear relations between the coils per phase and the thrust generated at similar power levels. The team settled on 4 coils, thus having a 4 pole induction motor as explicited by figure 57b. To determine the number of turns, a parametric sweep was performed while also deriving the peak voltage required to operate at the simulation's current of 50$[A]$. The settled-upon number was 40 turns, producing the force profile presented in figure 56a.



(a) Theorical Thrust of the New Induction Motor at 50$[A]$



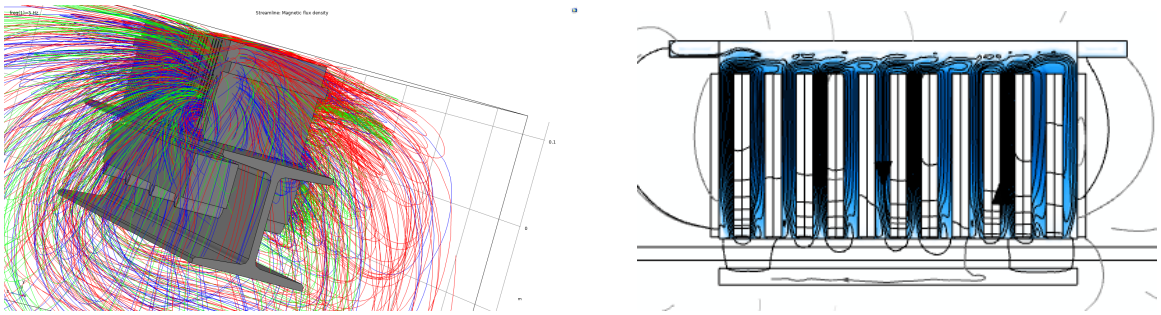
(b) Phase Configuration of the Prototype

Fig. 56: Windings and Power Production

5.2.2 Mass & Geometry

The mass constraint directly imposes a size constraint. To optimize the available volume, the first parameter set was the motor's width. A prior flux density COMSOL analysis revealed the majority of magnetic field flux losses of the core where located on the sides of the asynchronous motor, as demonstrated in figure 57a. Hence, the width ought to be slightly lower than the rail's width so that stray flux may penetrate the aluminum and

contribute to the eddy current formation while ensuring the maximal area covered by the induced current.



(a) Field lines Simulation of the First Motor (b) Prototype's 2D flux lines

Fig. 57: Windings and Power Production

The length of the newest prototype was optimized to reduce to weight of the electric motor while allowing the winding configurations presented in 5.2.1 with a maximizing input current of 120$[A]$. The estimated mass, including a set of two back-irons to increase the motor's efficiency, is to be around 22$[kg]$. With the resulting geometry, the expected generated thrust is 30$[N]$ at the simulated current of 50$[A]$.

References

Standard Test Methods for Tension Testing of Metallic Materials (Test Methods No. E8/E8M 22). (2022, July). American Association State. United States. https://compass.astm.org/document/?contentCode=ASTM%7CE0008_E0008M-22%7Cen-US&proxycl=https%3A%2F%2Fsecure.astm.org&fromLogin=true

Timperio, C. (2018). Linear induction motor (lim) for hyperloop pod prototypes. *ETH Zürich*.

A Appendix - Design

A.1 Aeroshell appendices

A.1.1 Metric display of the mesh quality

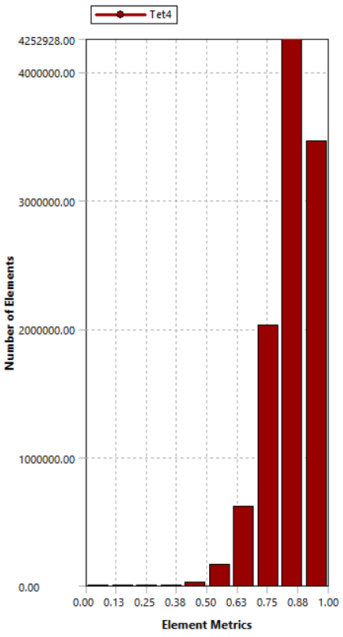


Fig. 58: Element Quality

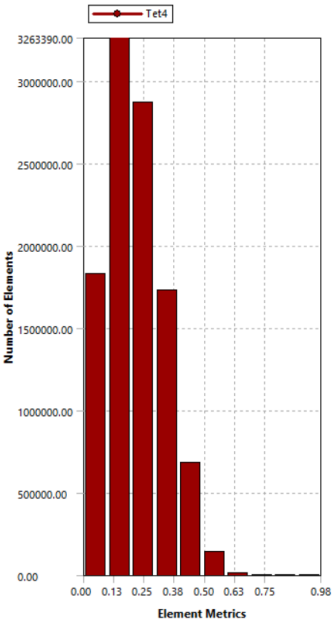


Fig. 59: Skewness

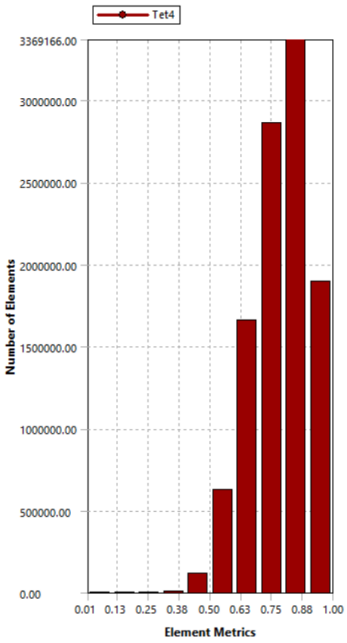


Fig. 60: Orthogonal Quality

A.1.2 y+ plot on the pod

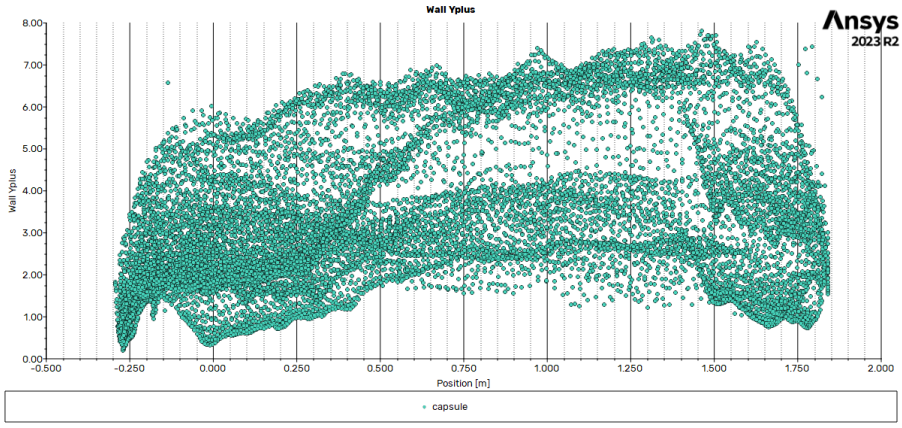


Fig. 61: y+ plot on the pod

A.1.3 y+ plot on the walls

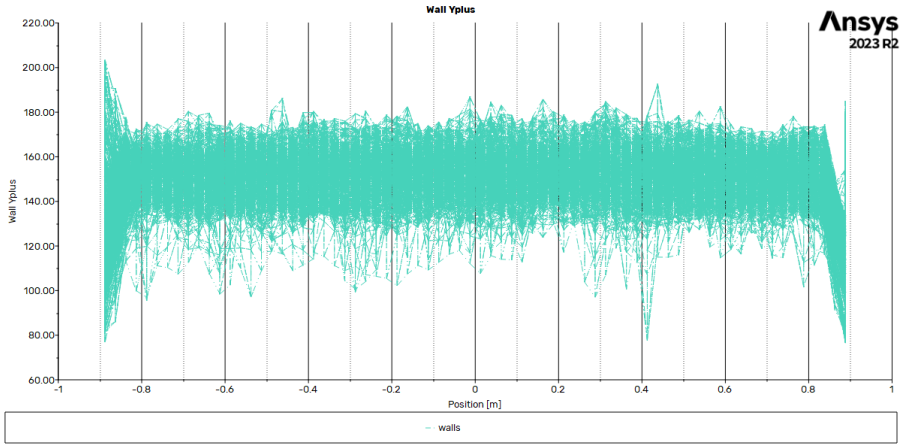


Fig. 62: y+ plot on the walls

A.1.4 Scaled residuals of the $k-\epsilon$ turbulent model used to obtain aerodynamics properties of the pod

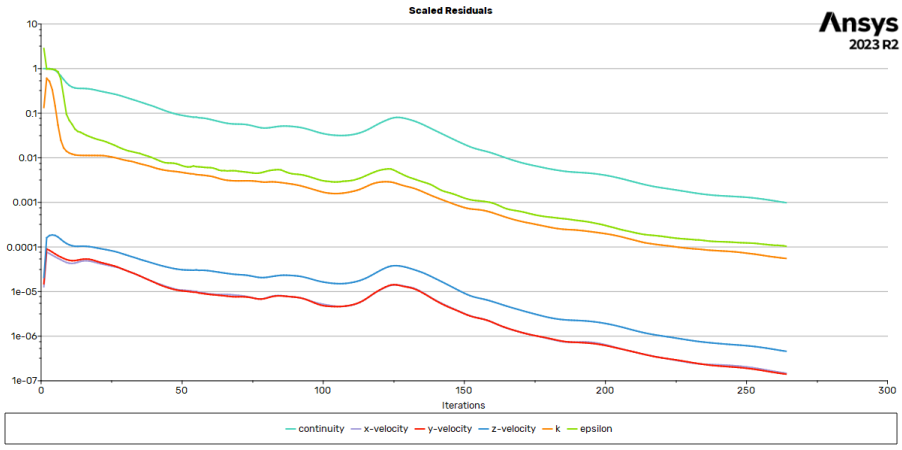


Fig. 63: Scaled residuals of the $k-\epsilon$ turbulent model

A.1.5 Plot of the drag coefficient over iteration

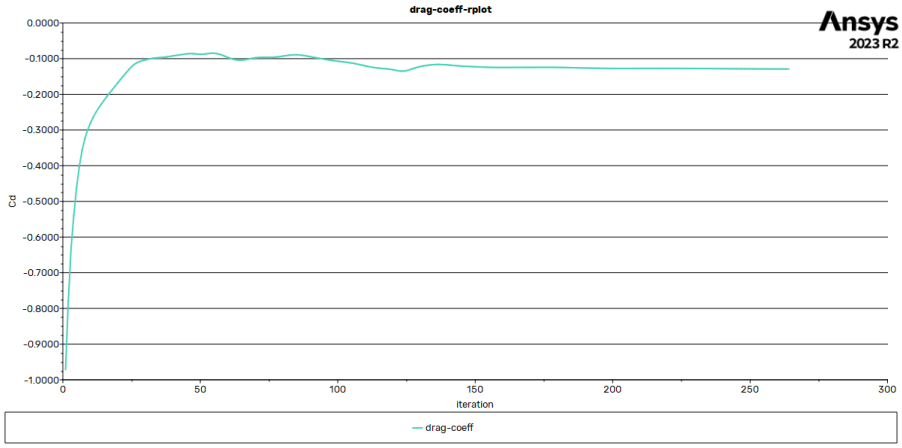


Fig. 64: Plot of the drag coefficient over iteration

A.1.6 Plot of the drag force over iteration

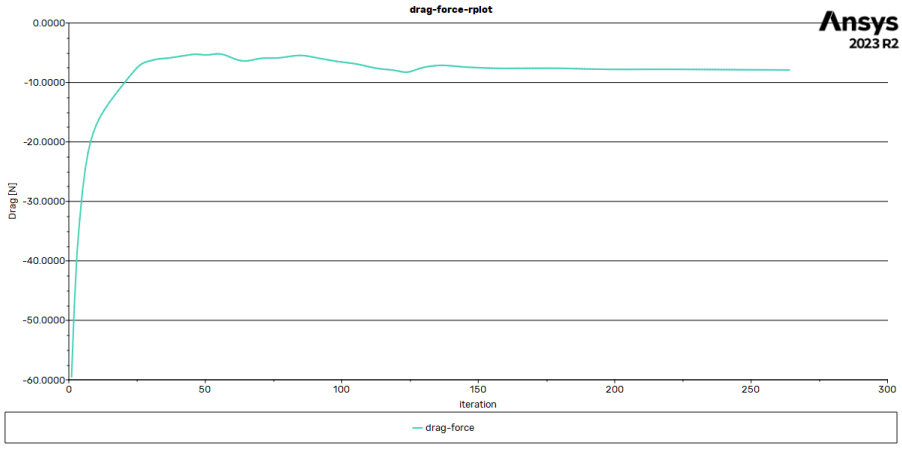


Fig. 65: Plot of the drag force over iteration

A.1.7 Plot of the lift force over iteration

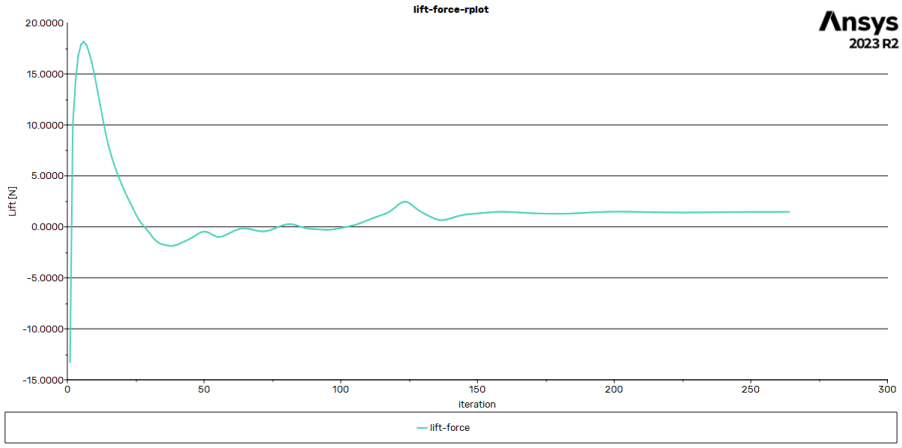


Fig. 66: Plot of the lift force over iteration

A.1.8 Plot of the lift coefficient over iteration

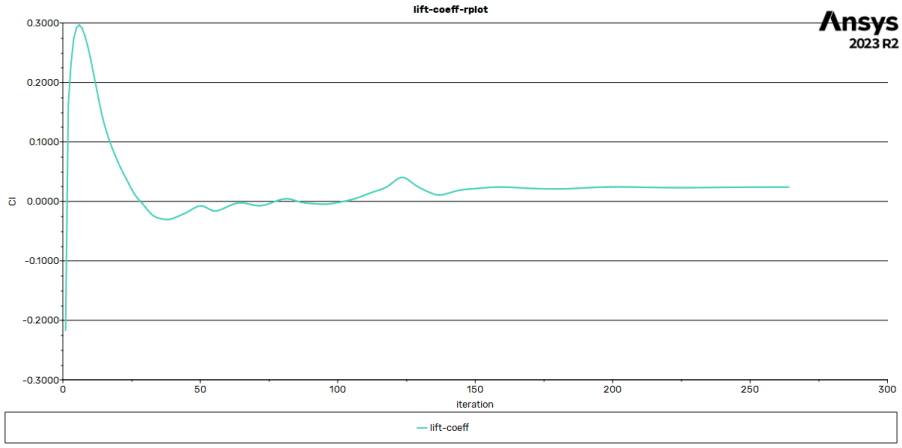


Fig. 67: Plot of the lift coefficient over iteration

A.2 Structures/Chassis appendices

A.2.1 Result of the structure modal analysis

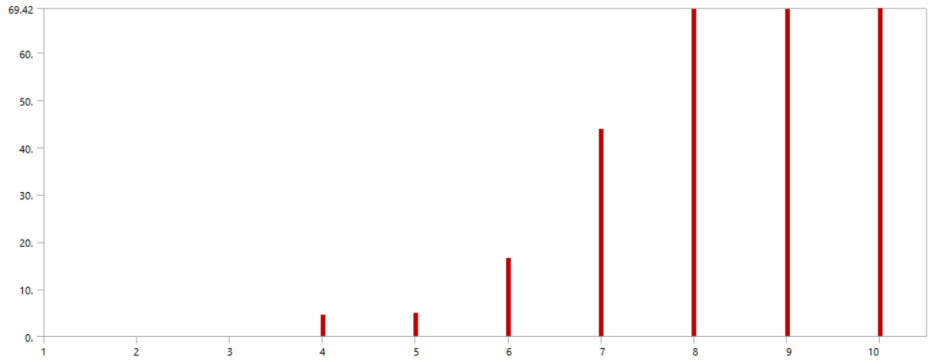


Fig. 68: 10 first natural frequency of the structure over a modal analysis (Hz)

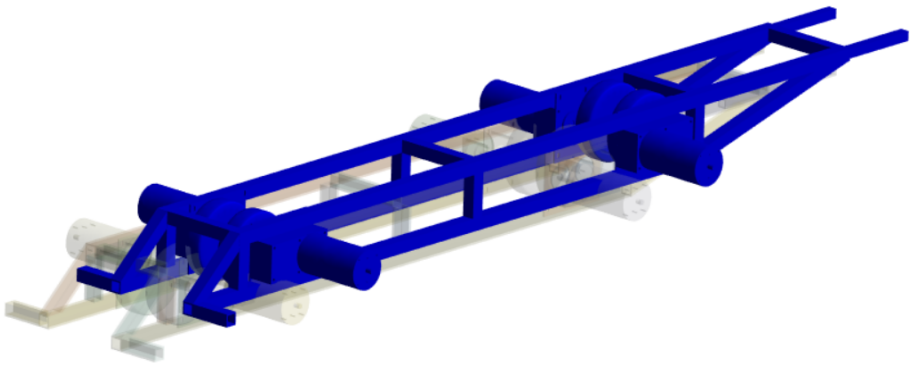


Fig. 69: Deformation at $0\,Hz$ (auto scale)

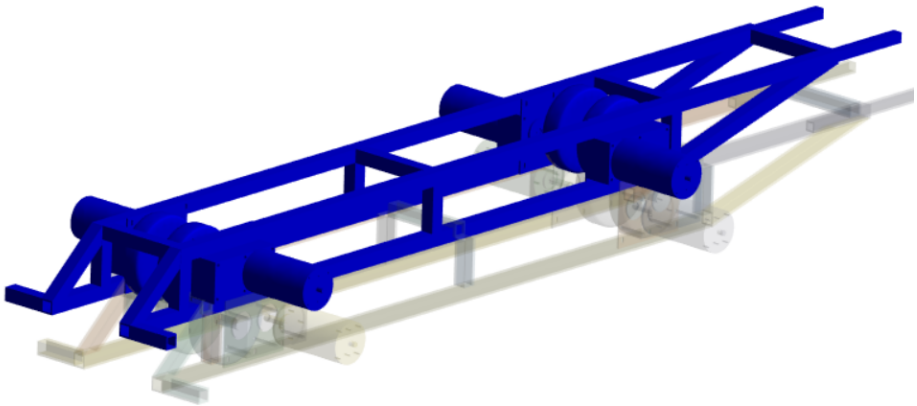


Fig. 70: Deformation at $0\,Hz$ (auto scale)

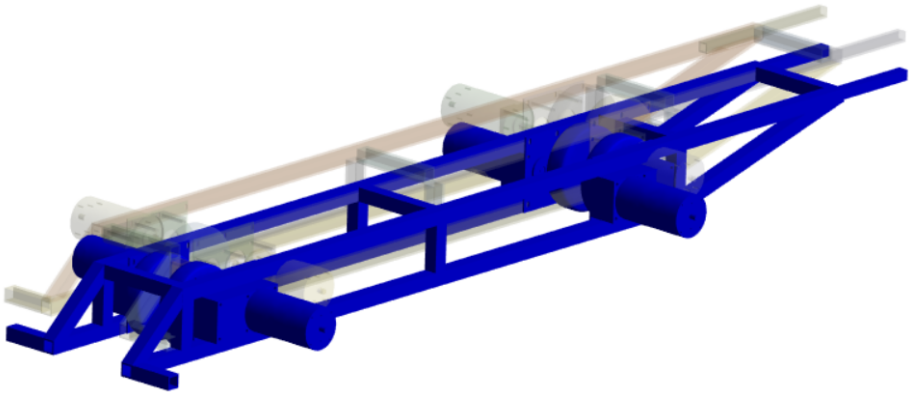


Fig. 71: Deformation at $2.4457e-004\,Hz$ (auto scale)

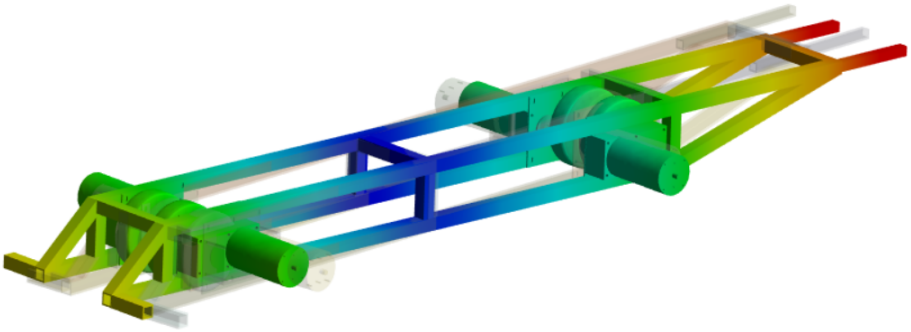


Fig. 72: Deformation at 4.4793 Hz (auto scale)

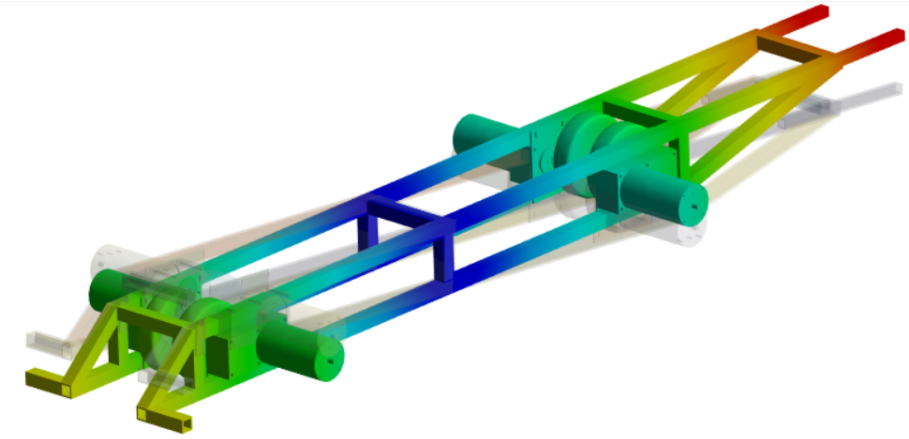


Fig. 73: Deformation at 4.9193 Hz (auto scale)

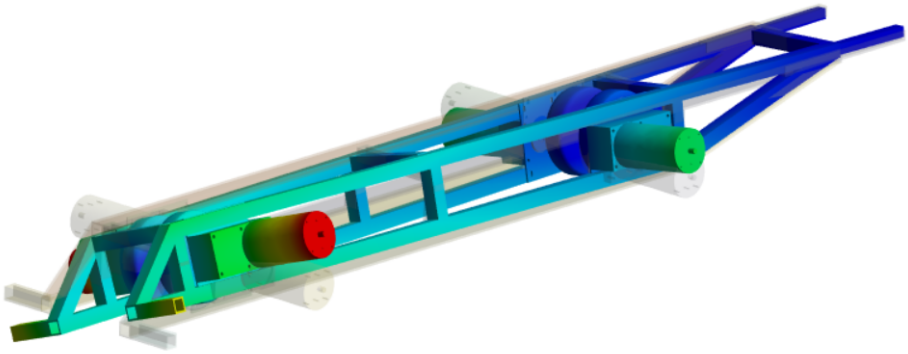


Fig. 74: Deformation at 16.612 Hz (auto scale)

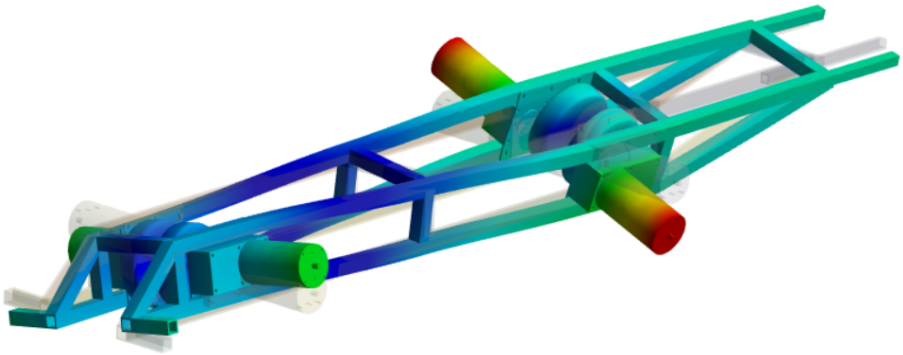


Fig. 75: Deformation at 43.835 Hz (auto scale)

A.3 Propulsion appendices

A.3.1 E30-400-48-G4 manifacturer's performance table

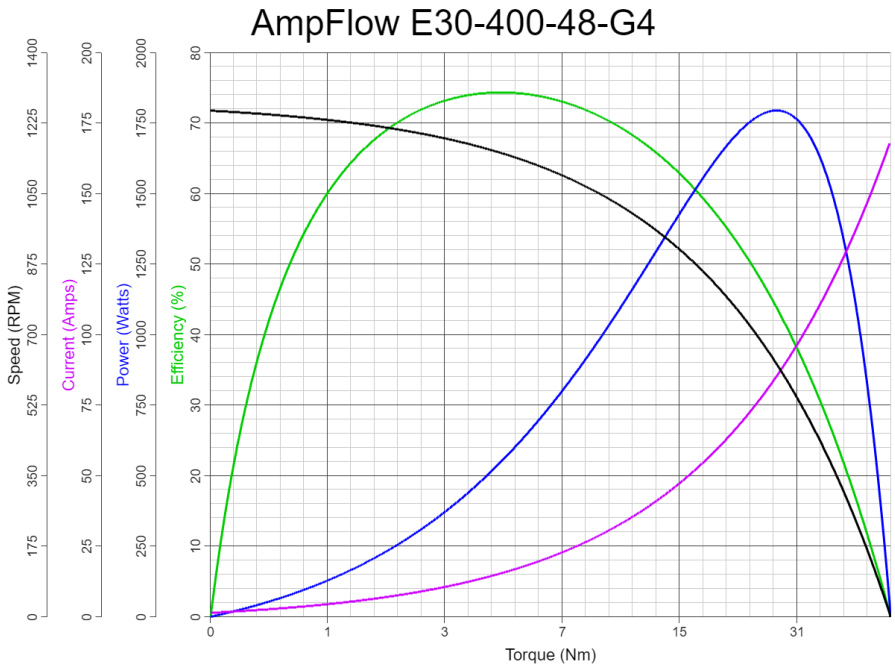


Fig. 76: E30-400-48-G4 manifacturer's performance table

A.3.2 Energy calculation for the propulsion needed

Energy calculation

To calculate the moment necessary to accelerate the capsule, it is necessary to use the method of Newton DCL, DCE on one of the wheels.

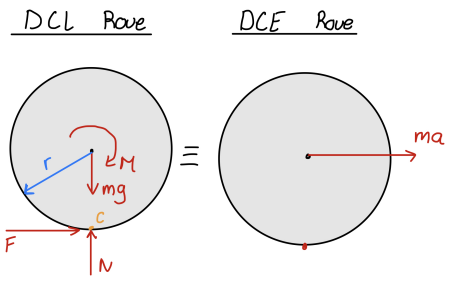


Fig. 77: DCL, DCE of one wheel

This way, by summing the moments on the instantaneous center of zero speed (ICZV),

we can determine the acceleration of the wheel.

$$\begin{aligned}\sum M_C = -M &= -m \cdot a \cdot r \\ 4.77 &= 25 \cdot a \cdot \frac{0.1524}{2} \\ a &= 2.5\end{aligned}$$

$a =$ acceleration $\left[\frac{m}{s^2}\right]$ $m =$ total mass on the wheel $[kg]$ $r =$ radius $[m]$

We know that the angular speed of the motor for a current of 16 Amps is $1150\,RPM$, we use this to calculate the nominal speed of the wheel.

$$\begin{aligned}v_{Roue} &= \omega \times R_{Rour} \\ &= \frac{1150 \cdot 2\pi}{60} \times \frac{0.1524}{2} \\ &= 9.17659\end{aligned}$$

$v_{Roue} =$ Velocity of the wheel $\left[\frac{m}{s}\right]$ $\omega =$ wheel rotation speed $\left[\frac{RAD}{s}\right]$ $R_{Rour} =$ radius of the wheel $[m]$

We find the time required by the capsule to reach this speed assuming that the acceleration is constant by the following relation;

$$\begin{aligned}a &= \frac{dv}{dt} \\ \int_0^t a dt' &= \int_0^v dv' \\ \int_0^t 2.5\ dt' &= \int_0^{9.18} dv' \\ 2.5\ t &= 9.18 \\ t &= 3.672\,\text{seconds}\end{aligned}$$

We now calculate the power needed to accomplish this acceleration by the relation of

the conservation of energy, neglecting the rotational kinetic energy of the wheel.

$$\begin{aligned} \text{Energy} \rightarrow P \cdot t &= \frac{1}{2} \cdot m \cdot v^2 \\ P &= \frac{m \cdot v^2}{2 \cdot t} \\ &= \frac{100 \cdot 9.18^2}{2 \cdot 3.627} \\ &= 1\,161\, Watt \end{aligned}$$

The theoretical power of an E30-400-48-G4 motor is $573\, Watt$. So, having 4 motors, the total power supplied is calculated:

$$\begin{aligned} P &= 573 \cdot 4 \\ &= 2292\, Watt \end{aligned}$$

Thus the power supplied is greater than the power required to accelerate the capsule to $9.18 m/s$, the E-30-400-48-G4 model is adequate.

A.4 Braking appendices

A.4.1 Analysis of the Hallback array

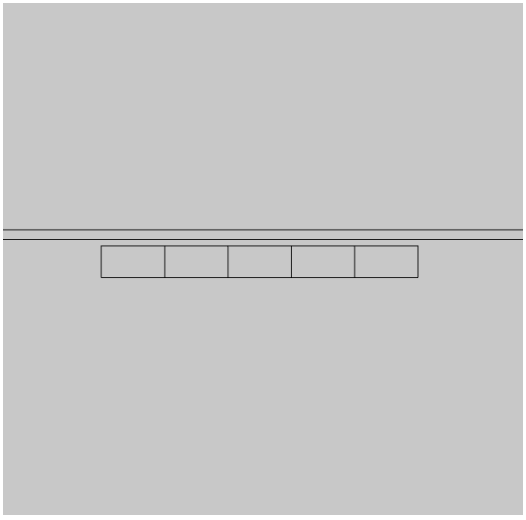


Fig. 78: Geometry used to simulate the effect of the Hallback array

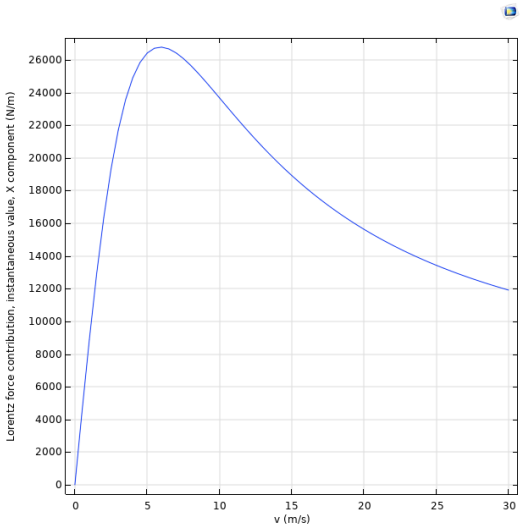


Fig. 79: Graph of the results of the simulation. The Lorentz force contribution is the braking force generated by the Hallback array. It is in function of the velocity v (m/s)

A.4.2 Acceleration compute from Newton second law

Calculation

We supose a constant mass m, constant forces and the momentum is define as $\vec{G} = m\vec{v}$.

$$\begin{aligned}\sum \vec{F} &= m\vec{a} = m\dot{\vec{v}} \\ &= m\frac{dv}{dt} = \frac{dm\vec{v}}{dt} \\ &= \frac{d\vec{G}}{dt}\end{aligned}$$

Then we distribute the dt and we integrate on both sides.

$$\begin{aligned}\sum \vec{F}\,dt &= d\vec{G} \\ \int_{t_0}^{t_1} \sum \vec{F}\,dt &= \int_{\vec{G}_0}^{\vec{G}_1} d\vec{G} \\ \sum \vec{F}(t_1 - t_0) &= \vec{G}_1 - \vec{G}_0 \\ \sum \vec{F} &= \frac{\vec{G}_1 - \vec{G}_0}{t_1 - t_0} \\ &= \frac{\Delta\vec{G}}{\Delta t}\end{aligned}$$

Where $\Delta\vec{G}$ is the change of momentum and Δt is the elapsed time. With this formula, we can compute the deceleration time knowing the masse of the pod m, the max velocity $\vec{v}_0$, and the total friction force.

Assuming total power failure, we sum the vector forces in the direction of the pod displacement $\vec{x}$. The only force acting is the friction force because the motors are in failure. With an initial speed $\vec{v}_0 = 9.14\,\vec{x}\,m/s$ and a total friction force $F_f = 159.052\,N$

$$\begin{aligned}\sum F|_x = -F_f &= \frac{\Delta G_x}{\Delta t} \\ \Delta t &= \frac{m\cancelto{0}{v_{1x}} - mv_{0x}}{-F_f} \\ &= \frac{-100\,[kg]\cdot 9.14\,[m/s]}{-159.052} \\ &= 5.7464\,s\end{aligned}$$

There the deceleration is $a = \frac{0-9.14}{5.7464} = -1.59\,m/s^2$.

A.4.3 Determination of the distance travelled in case of total power failure using only the speed reducer friction force

We hypothesize that the deceleration is constant and the only pod displacement direction is $\vec{x}$. The speed and deceleration could be describe as the rate of change of the over time such as;

$$v = \frac{ds}{dt} \qquad a = \frac{dv}{dt}$$

Rearrange the termes to obtain:

$$dv = adt \rightarrow \int_{v_0}^{v_1} dv = a\int_{t_0}^{t_1} dt \rightarrow v_1 = v_0 + a(t_1 - t_0) \tag{i}$$

$$ds = vdt \rightarrow \int_{s_0}^{s_1} ds = \int_{t_0}^{t_1} \underbrace{(v_0 + a(t_1 - t_0))}_{\text{Eq}\,i}\,dt \rightarrow s_1 - s_0 = v_0 t + \frac{at^2}{2}$$

So the distance travelled is $\Delta s = 9.14 \cdot 22.3 + \frac{0.409 \cdot 22.3^2}{2} = 305.5\,m$

A.5 Electronics, Controls, and Communication appendices

A.5.1 Template Base station

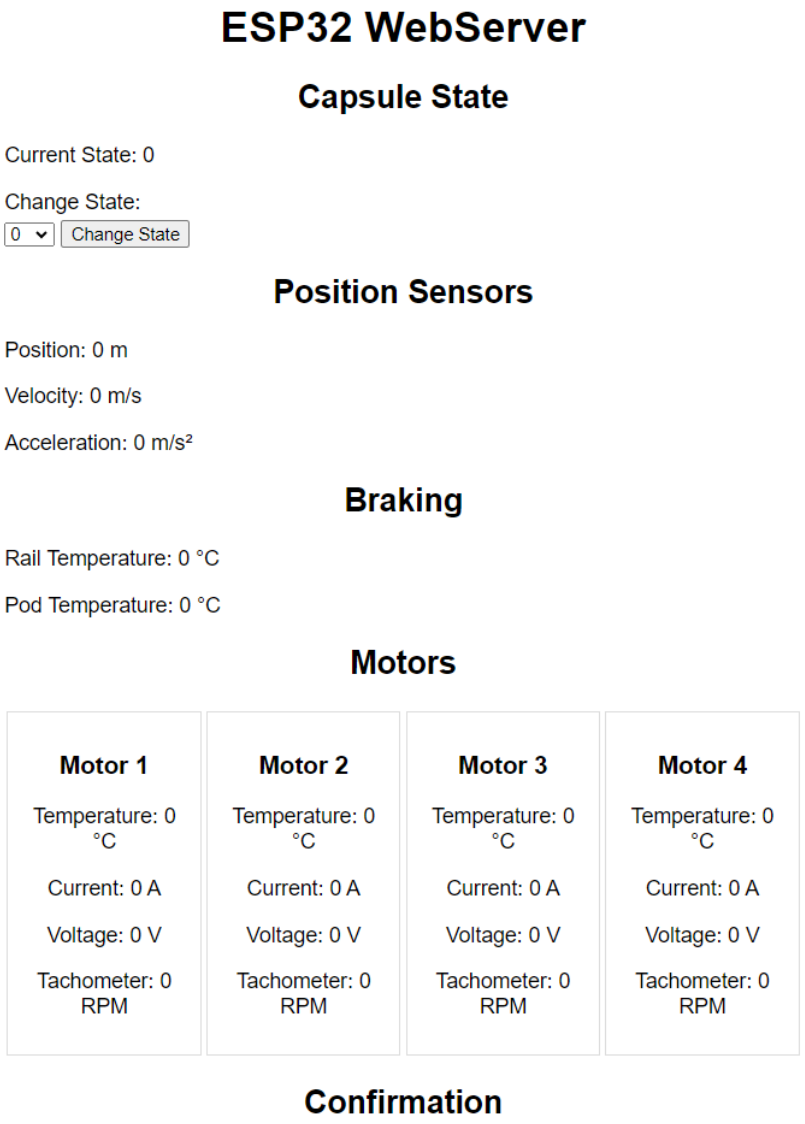


Fig. 80: Template Base station

A.5.2 State Machine

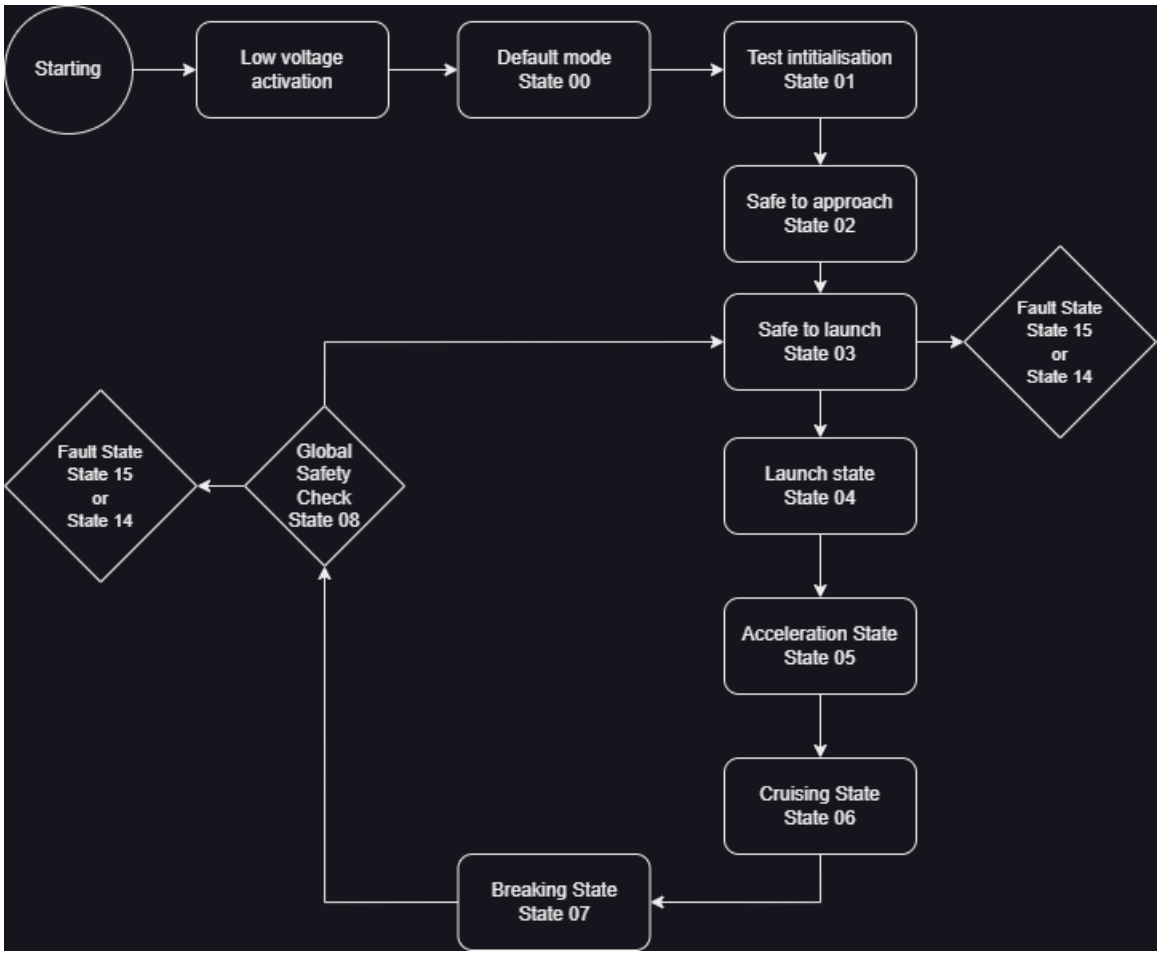


Fig. 81: State Machine

A.5.3 Full State Machine

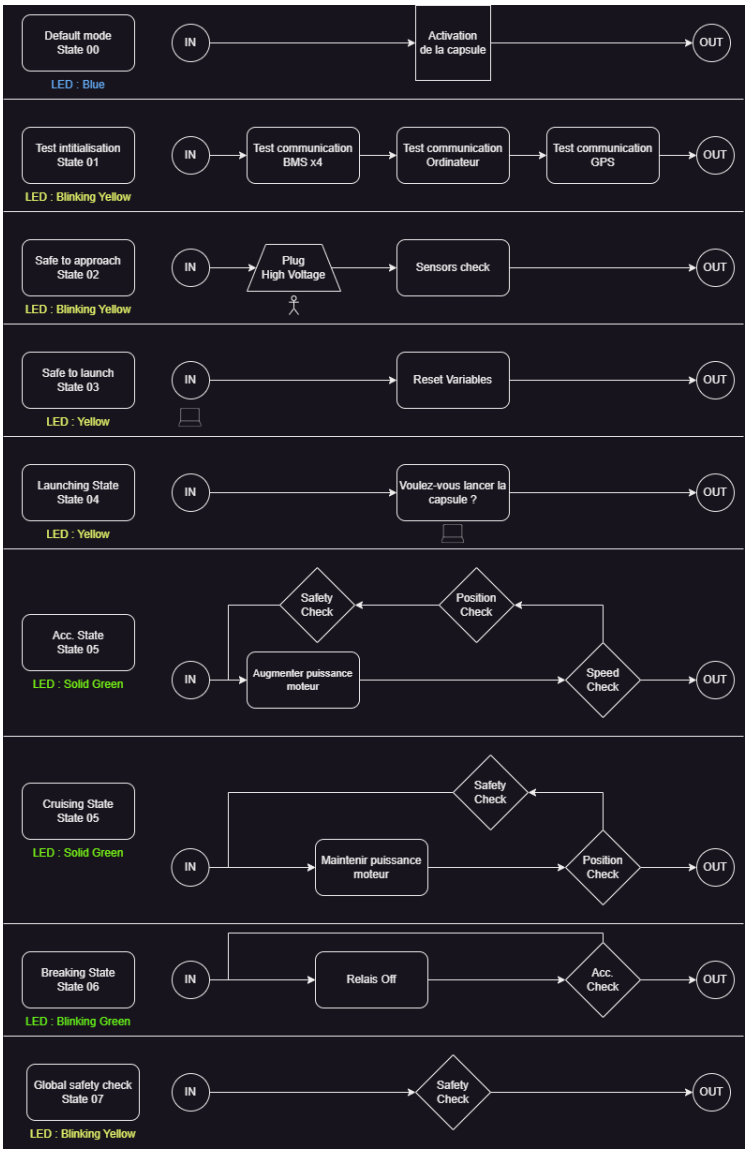
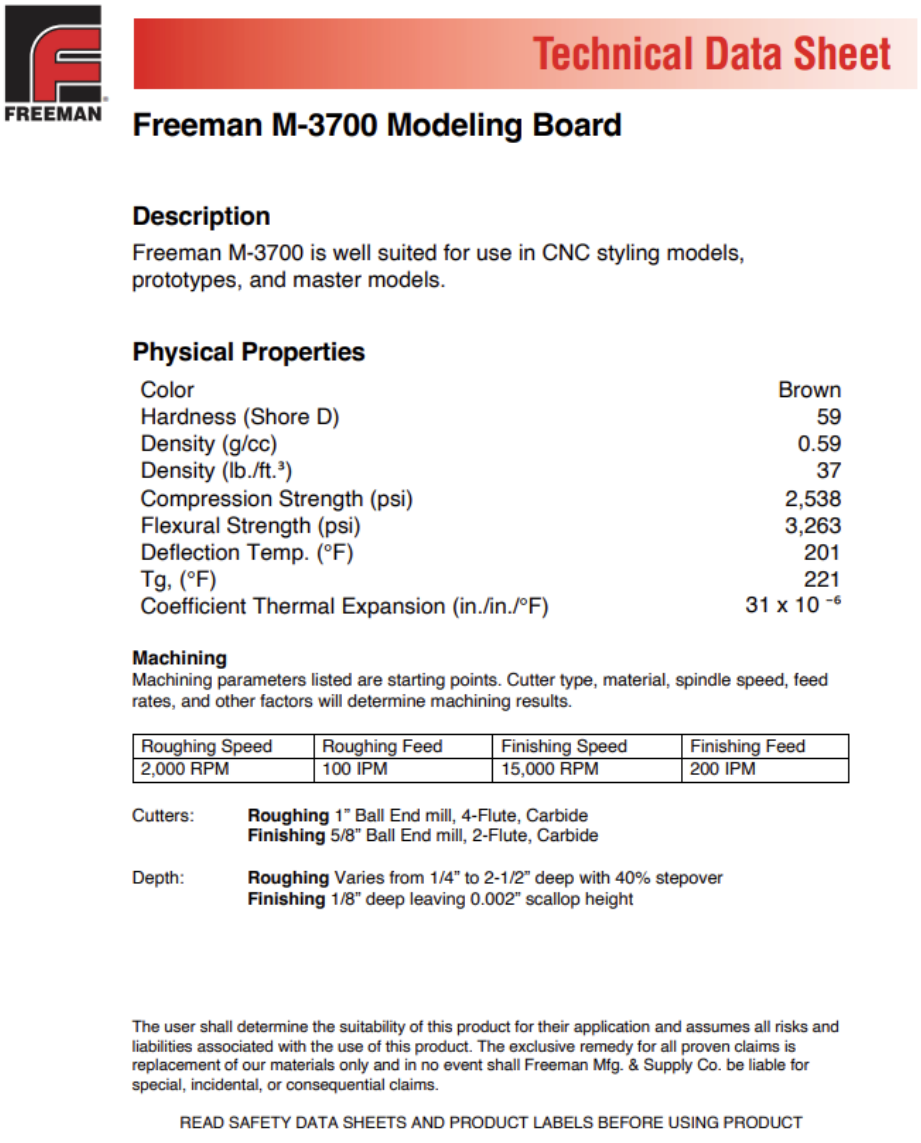


Fig. 82: Full State Machine

B Appendix - Manifacturing

B.1 Aeroshell appendices

B.1.1 Freeman M-3700 Modeling Board technical data sheet



FREEMAN

Technical Data Sheet

Freeman M-3700 Modeling Board

Description

Freeman M-3700 is well suited for use in CNC styling models, prototypes, and master models.

Physical Properties

Color	Brown
Hardness (Shore D)	59
Density (g/cc)	0.59
Density (lb./ft.^{3})	37
Compression Strength (psi)	2,538
Flexural Strength (psi)	3,263
Deflection Temp. (°F)	201
Tg, (°F)	221
Coefficient Thermal Expansion (in./in./°F)	31 x 10 $^{-6}$

Machining

Machining parameters listed are starting points. Cutter type, material, spindle speed, feed rates, and other factors will determine machining results.

Roughing Speed	Roughing Feed	Finishing Speed	Finishing Feed
2,000 RPM	100 IPM	15,000 RPM	200 IPM

Cutters: **Roughing** 1" Ball End mill, 4-Flute, Carbide
Finishing 5/8" Ball End mill, 2-Flute, Carbide

Depth: **Roughing** Varies from 1/4" to 2-1/2" deep with 40% stepover
Finishing 1/8" deep leaving 0.002" scallop height

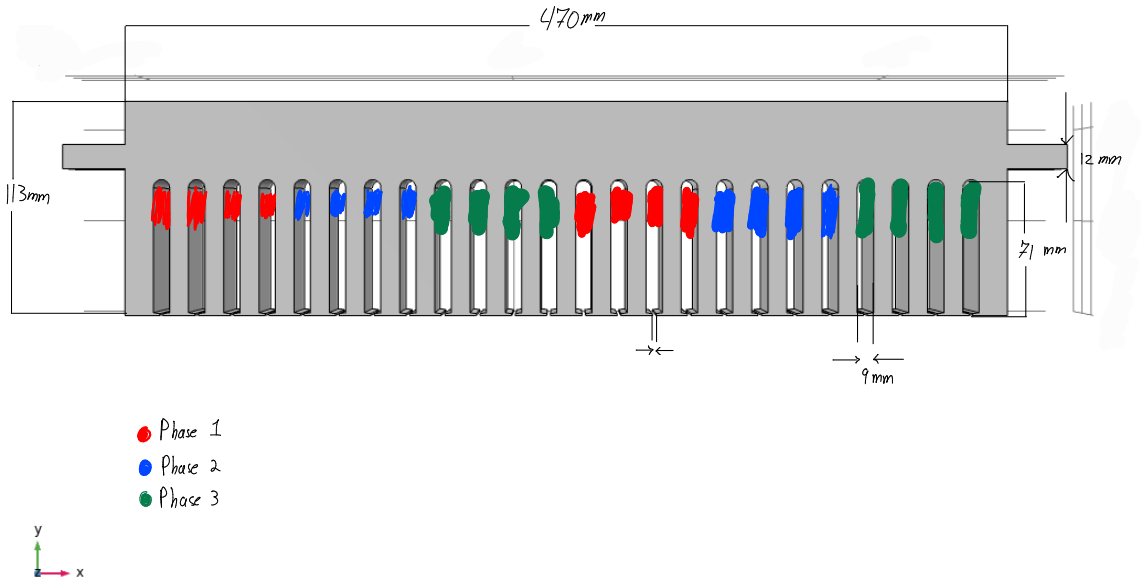
The user shall determine the suitability of this product for their application and assumes all risks and liabilities associated with the use of this product. The exclusive remedy for all proven claims is replacement of our materials only and in no event shall Freeman Mfg. & Supply Co. be liable for special, incidental, or consequential claims.

READ SAFETY DATA SHEETS AND PRODUCT LABELS BEFORE USING PRODUCT

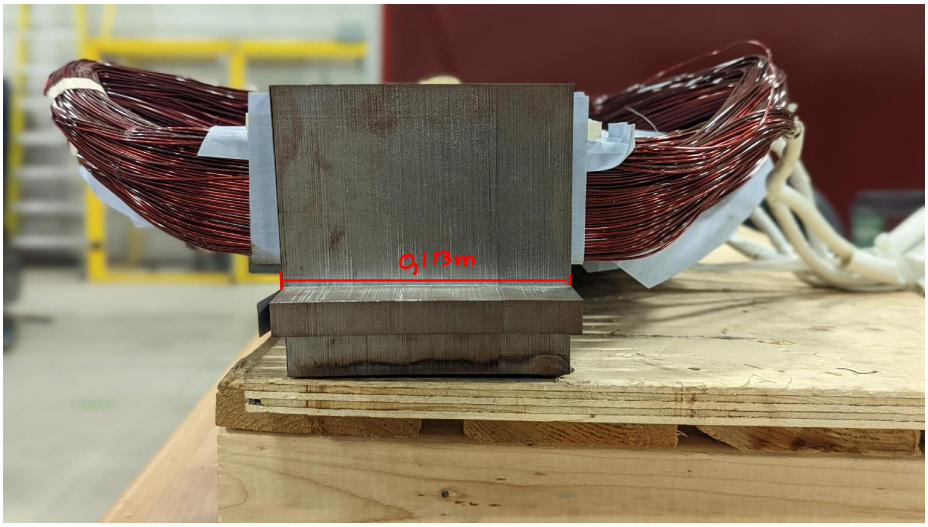
Fig. 83: Caption

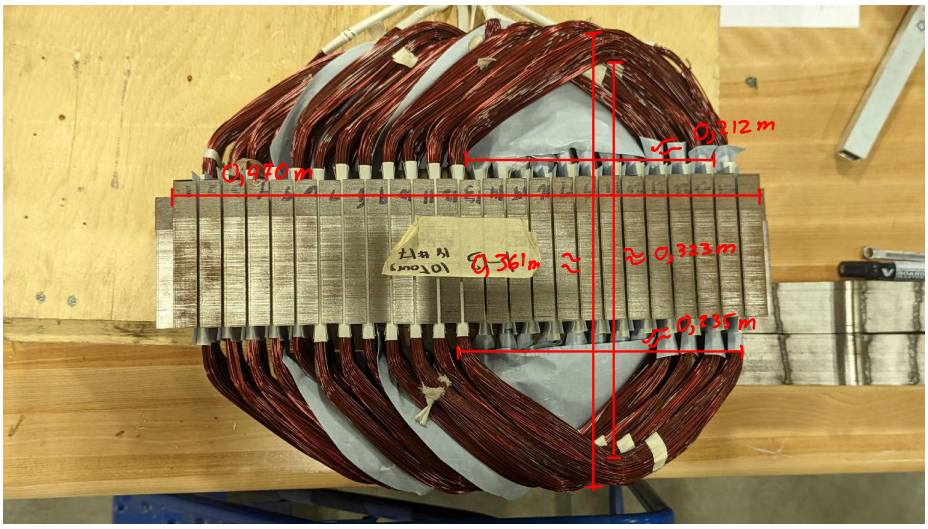
C Appendix - Linear Electromagnetical Propulsion Project

C.0.1 LIM Informations



0,113m
0,012m
0,47m


0,153m


0,212 m
0,470 m
0,361 m
≈
≈ 0,323 m
0,235 m

C.0.2 Simcenter Motorsolve Report

Simcenter Motorsolve Report

Material

Name:
M-19 24 Ga

Database:
Non-Oriented AISI Silicon Steel Materials

Color:
ARGB = (255, 129, 115, 168)

Description:

Lamination thickness: 0.64 mm (0.025 in)

Equivalent: EN 10106 (1995) M350-65A
Equivalent: IEC 60404-8-4 (1998) M350-65A5
Equivalent: ASTM A677 (1999) 64F210

Categories:

1. Core material

Classifications:
None

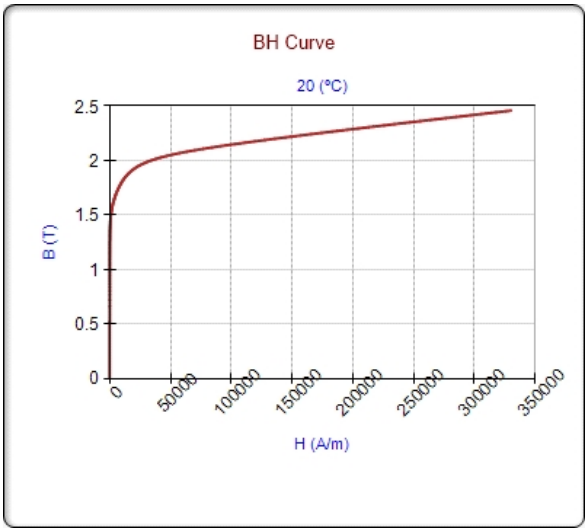
Properties:

- Magnetic Permeability
- Iron Loss
- Electric Resistivity
- Electric Permittivity
- Thermal Conductivity
- Thermal Specific Heat Capacity
- Mass Density
- Lamination Thickness

Magnetic Permeability
Type: Nonlinear, Isotropic, Real

Temperature: 20 (°C)

H (A/m)	B (T)
0	0
49.1195539	0.518749149
62.75609992	0.661362733
68.79453615	0.720435214
73.71730499	0.765763123
78.12986982	0.803976318
82.26776131	0.837642819
86.25440395	0.868079662
90.16632455	0.896069228
94.05689561	0.92212128
97.96693171	0.946589903
101.9301186	0.969732902
105.9761075	0.991744899
110.13245	1.012777098
114.4259191	1.032949741
118.8834958	1.052360322
123.5331765	1.071089188
128.4046917	1.089203488
133.5302006	1.106760022
138.9450031	1.123807346
144.6883074	1.140387353
150.8040848	1.156536489
157.342045	1.172286696
164.3587652	1.187666154
171.9190088	1.202699874
180.0972743	1.217410175
188.9796146	1.231817073
198.6657699	1.245938601
209.271651	1.259791073
220.9322001	1.273389307
233.8046308	1.286746807
248.072005	1.299875923
263.9470328	1.312787984
281.6758778	1.325493412
301.5416025	1.338001821
323.8667231	1.350322101
349.0141851	1.362462495
377.3860044	1.374430658
409.4189657	1.386233719
445.577243	1.397878328
486.3426628	1.4093707
532.2043781	1.420716655
583.6505347	1.43192165
641.1644976	1.442990812
705.2270167	1.453928964
776.3235929	1.46474065
854.954079	1.475430159
941.6401525	1.486001541
1036.926234	1.496458628
1141.370617	1.506805048
1255.525588	1.517044242
1379.907758	1.527179478
1514.962274	1.537213858



1661.026618	1.547150334
1818.300727	1.556991718
1986.829623	1.566740688
2166.502407	1.576399801
2357.068046	1.585971495
2558.164798	1.595458102
2769.357777	1.604861853
2990.178496	1.614184879
3220.161226	1.623429227
3458.872838	1.632596853
3705.934833	1.641689639
3961.037768	1.650709386
4223.949282	1.659657827
4494.517228	1.668536628
4772.669486	1.677347389
5058.411723	1.686091651
5351.824141	1.694770896
5653.057929	1.703386556
5962.331907	1.711940005
6279.929663	1.720432574
6606.197366	1.728865544
6941.542341	1.737240152
7286.432442	1.745557593
7641.39623	1.753819023
8007.023942	1.762025557
8383.969246	1.770178277
8772.951759	1.778278226
9174.760318	1.786326417
9590.257016	1.794323829
10020.38198	1.802271412
10466.15893	1.810170085
10928.70151	1.818020741
11409.22043	1.825824244
11909.03137	1.833581434
12429.56382	1.841293127
12972.37066	1.848960112
13539.13858	1.85658316
14131.69939	1.864163017
14752.04202	1.871700409
15402.32517	1.879196041
16084.89059	1.8866506
16802.27654	1.894064755
17557.23138	1.901439154
18352.72672	1.908774431
19191.96963	1.916071202
20078.41323	1.923330066
21015.76468	1.93055161
22007.98961	1.937736401
23059.31143	1.944884997
24174.20412	1.951997938
25357.37672	1.959075753
26613.74745	1.966118957
27948.40574	1.973128053
29366.56021	1.980103531
30873.47128	1.987045871

32474.36766	1.993955541
34174.347	2.000832997
35978.26233	2.007678687
37890.59767	2.014493045
39915.33761	2.021276499
42055.83777	2.028029465
44314.7039	2.03475235
46693.68803	2.041445554
49193.61	2.048109465
51814.31076	2.054744465
54554.64188	2.061350927
57412.49251	2.067929216
60384.85181	2.074479689
63467.90184	2.081002697
66657.13383	2.087498582
69947.4792	2.093967679
73333.44683	2.100410318
76809.25848	2.106826819
80368.97591	2.1132175
84006.61491	2.119582668
87716.24319	2.125922628
91492.06107	2.132237675
95328.46469	2.138528103
99220.09326	2.144794196
103161.8616	2.151036234
107148.9803	2.157254493
111176.966	2.163449243
115241.6424	2.169620747
119339.1362	2.175769266
123465.8678	2.181895054
127618.538	2.187998362
131794.1139	2.194079436
135989.8125	2.200138517
140203.0841	2.206175841
144431.5954	2.212191641
148673.2136	2.218186147
152925.99	2.224159582
157188.1455	2.230112167
161458.0559	2.236044119
165734.2395	2.241955651
170015.3441	2.247846972
174300.136	2.253718289
178587.4895	2.259569803
182876.3777	2.265401715
187165.8633	2.271214219
191455.0909	2.277007507
195743.2798	2.28278177
200029.7173	2.288537194
204313.7527	2.294273961
208594.7922	2.299992252
212872.2934	2.305692245
217145.7611	2.311374113
221414.7432	2.317038029
225678.8273	2.322684162
229937.6368	2.328312678

234190.828	2.333923741
238438.0877	2.339517513
242679.1303	2.345094152
246913.6955	2.350653815
251141.5464	2.356196655
255362.4676	2.361722826
259576.2631	2.367232475
263782.7551	2.372725751
267981.7827	2.378202799
272173.1998	2.383663761
276356.8749	2.389108779
280532.6891	2.394537991
284700.5358	2.399951535
288860.3191	2.405349544
293011.9536	2.410732152
297155.3631	2.416099491
301290.4804	2.421451688
305417.246	2.426788873
309535.608	2.43211117
313645.5213	2.437418703
317746.9473	2.442711595
321839.8532	2.447989965
325924.2115	2.453253933
330000	2.458503617

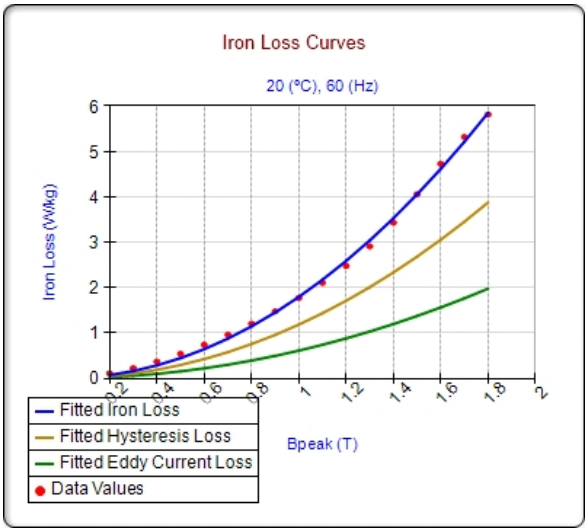
Iron Loss
Type: Nonlinear, Isotropic, Real

Steinmetz equation coefficients at 20 (°C):
Kh = 0.0197195053209291
α = 1
β = 2.01766709449797
Ke = 0.000169372798838872

Method: Absolute Error

Temperature: 20 (°C), Frequency: 60 (Hz)

Bpeak (T)	Iron Loss (W/kg)
0.2	0.108026508
0.3	0.222666885
0.4	0.368171978
0.5	0.540132542
0.6	0.738548578
0.7	0.959010841
0.8	1.205928574
0.9	1.474892534
1	1.772516588
1.1	2.103209981
1.2	2.475791204
1.3	2.907897238
1.4	3.4303928
1.5	4.056505624
1.6	4.728915524
1.7	5.319754387
1.8	5.815794476



Electric Resistivity
Type: Linear, Isotropic, Real

Temperature (°C)	Resisitivity (Ωm)
20	5.2E-07

Electric Permittivity
Type: Linear, Isotropic, Real

Temperature (°C)	Relative Permittivity
20	1

Thermal Conductivity
Type: Linear, Isotropic, Real

Temperature (°C)	Conductivity (W/m°C)
20	25

Thermal Specific Heat Capacity
Type: Linear, Isotropic, Real

Temperature (°C)	Specific Heat Capacity (J/kg°C)
20	490

Mass Density
Type: Linear, Isotropic, Real

Temperature (°C)	Mass Density (kg/m^3)
20	7650

Lamination Thickness
0.0251969 (in)